The Higman–Sims incidence geometries: classification, constructions and corrected source data

Human Compression project

Abstract

We assemble complete finite proofs for the classification and two constructions in D. Leemans, *The residually weakly primitive geometries of HS*, together with the acquired supplementary diagrams. The classification counts are recovered. The diagrams, local groups and full automorphism groups are compared occurrence by occurrence, with corrections stated explicitly. Earlier sporadic classifications quoted by the article retain their own equivalence conventions and premises. The accompanying proof volume preserves the complete accepted arguments and finite certificates on which this synthesis depends.

1 The source and the mathematical statement

The fixed article is the six-page edition in Australasian Journal of Combinatorics **33** (2005), 231–236, denoted by L091. Its acquired supplement is the twenty-three-page edition dated 6 October 2003. The original results and constructions remain attributed to that article and its cited authors. Here an automorphism preserves each type; a correlation may permute the types. Isomorphism of geometries may also permute types. This convention is essential when counting the objects.

Let $G = \mathrm{HS}$ in the retained faithful permutation model on one hundred points, with $|G| = 44\,352\,000$. An eligible geometry is firm, residually connected and flag-transitive under G, residually weakly primitive, locally doubly transitive, and satisfies the geometric intersection condition $(\mathrm{IP})_2$. Weak primitivity means that at least one type in every nonempty residue is primitive. It does not assert primitivity on every type. The geometric condition $(\mathrm{IP})_2$ requires each rank-two residue to be a partial linear space or a generalized digon; it is distinct from an identity between subgroup intersections.

Theorem 1. *Under the explicit group and software inputs in Section 9, the eligible geometries have, up to incidence isomorphism, the following rank counts:*

rank	1	2	3	4	5	6	≥ 7
all geometries	1	6	30	41	8	1	0
not truncations	0	4	20	41	8	1	0.

The eighty-six classes of rank at least two correspond to the eighty-six supplementary diagrams after the explicit corrections recorded below. Their local data, full groups, thirteen printed truncation links, conjugacy notes and three residual-primitivity annotations have complete finite certificates. Five printed doubling links are verified; three are incompatible with the precise two-point operation used by the source adapter. Both Hoffman–Singleton constructions in the article are realized by explicit incidence isomorphisms.

The theorem concerns a corrected source disposition. It does not say that every printed expression is correct. In particular the original classification counts survive, while several subsidiary diagrams and group labels change. The separate comparison with other sporadic groups is stated in Section 8; its borrowed classification premises are not assertions proved by the HS census.

2 Coset recognition and complete enumeration

We recall the common argument because it identifies exactly what the finite enumeration proves. Suppose a group acts on the right, type-preservingly and flag-transitively, and choose a base chamber $(x_i)_{i\in I}$. Set $H_i = G_{x_i}$. The map

$$H_i g \longmapsto x_i^g$$

is a bijection on each type: two images agree precisely when their right cosets agree. Cosets meeting at g give incident translates of base objects. Conversely, flag transitivity on each pair of types gives a common representative for every incident pair. Thus this map preserves incidence and nonincidence.

For a chamber stabilizer B, let P_i be its panel stabilizers and put $K_\varnothing = B$, $K_S = \langle P_i : i \in S\rangle$. The recognition conditions are

$$K_I = G, \qquad K_S \cap K_T = K_{S\cap T}, \qquad [P_i : B] \geq 2, \tag{1}$$

together with typed coset Helly for the maximal parabolics $H_i = K_{I\setminus\{i\}}$. Typed Helly means that any pairwise intersecting family of cosets of distinct types has a common element. It is not inferred from the subgroup identity in (1).

Here is the exact finite Helly criterion used in the common proof. For every i, $J \subseteq I \setminus \{i\}$ with $|J| \geq 2$, and chosen $j_0 \in J$, require that each $g \in H_{j_0}$ satisfying $g \in H_i H_j$ for all $j \in J$ admit $k \in \bigcap_{j\in J} H_j$ with $H_i g = H_i k$. Necessity follows by applying Helly to $H_i g$ and the base cosets H_j. Conversely, induct on the number of cosets in a pairwise intersecting family. Translate a common element of all but one to the identity, then choose a representative of the remaining coset in H_{j_0}. The criterion supplies a common representative for the whole family, completing the induction.

These conditions recognize the desired firm, connected coset geometry. Helly gives flag transitivity and identifies residues with the coset systems in K_S. Equation (1) identifies all base flag stabilizers. Panels have at least two objects. The subgroup generated by the parabolics of a residue has exactly the objects reachable by coset paths; their generation of K_S proves connectedness. Conversely, residual connectivity reconstructs every complementary flag stabilizer from its panels, by induction on $|S|$, giving (1). The remaining tests are literal local conditions: maximal proper subgroups for primitivity, two-transitive panel actions, and common neighbours for $(\mathrm{IP})_2$.

Lemma 2. *Given complete representatives of the subgroup classes of G and of the maximal subgroups inside each representative, maximal augmentation enumerates every eligible panel family.*

Proof. If a maximal inclusion $M < K$ is transported so that $M^c = H$, every maximal overgroup of H in that inclusion class is $(K^c)^n$ for some $n \in N_G(H)$. Indeed, compare any transporter of the inclusion with the chosen c; their quotient normalizes H. Conversely every displayed conjugate is such an inclusion. This gives the complete actual list of panels above each fixed B, retaining only the required local two-transitive actions.

For a target of rank r, weak primitivity supplies a maximal parabolic H in its acting group K. Deleting the complementary panel leaves an eligible residue with the same Borel. By induction its panel tuple appears at the previous level. The missing panel lies in K and outside H, and the complete overgroup list includes K. Taking representatives under the stabilizer of the old tuple loses no extension: equivalent choices are related by an element fixing that tuple. The recognition and local tests accept exactly the eligible extensions, after which normalizer and type orbits remove duplicates.

Adjoining a panel strictly enlarges the generated group, since otherwise (1) would force that panel to equal B. Thus a tuple gives a strict subgroup chain. Its rank is bounded by $\Omega([G : B])$, counting prime factors with multiplicity. The finite record must reach an empty next level; a chosen rank cutoff alone is not an exhaustion certificate. Finally let B range over the complete subgroup catalogue. □

The accepted catalogue has 589 subgroup classes and 3075 maximal inclusion rows. The terminal inventory covers all 589 anchors: 529 have completed enumerations, and 60 have complete proper-generation certificates. For the latter, the join of every admissible panel is proper in G, so no selected family can generate G. More generally, for a partial family T, every possible added panel lies among those compatible with every member of T. If the join of this entire compatible set is proper, no full-group extension exists. Conjugation preserves compatibility, so rows may be transported along normalizer orbits. Every prefix of a genuine full-group family survives these bounds. Consequently these exclusions preserve the induction of Lemma 2; they are not invented empty search levels.

The complete input has 130 families under inner conjugation and type permutation, with rank counts $(6, 44, 67, 12, 1)$. Its full-automorphism quotient has counts $(6, 30, 41, 8, 1)$. To compare two tuples, fix a target type permutation, successively transport each subgroup, and replace the remaining acting group by the stabilizer of the subgroups already placed. The possible completions form precisely a transporter coset, so this procedure loses no solution. Every positive comparison retains its actual permutation and checks the entire family. The full ambient automorphism action is realized by the acquired permutation normalizer of order 88 704 000, with trivial centralizer of G.

3 Forgetting the specified action and removing truncations

The preceding quotient is initially a quotient of actions. An arbitrary incidence isomorphism need not preserve the specified G, so a further argument is required. For each type subset J, flag transitivity gives

$$\#\{\text{flags of type } J\} = |G : G_J|, \qquad G_J = \bigcap_{j \in J} H_j.$$

These are intrinsic incidence counts. Under a type permutation their coordinates are permuted. Since the group order is fixed, the stabilizer order profiles also separate incidence geometries. On the eighty-six representatives they separate all but pairs $(14, 15)$ and $(83, 84)$. The complete profile over all type subsets separates 83 and 84.

For $14, 15$, the distinct type sizes are $231000, 23100, 1100$. Define two objects of the last type to be adjacent when some object of the first type is incident with both. The resulting intrinsic graphs have 1100 vertices and valency 105. Blocks are determined by their sixteen-point shadows: the shadow stabilizer has exactly the block stabilizer's order 1920. Lines have repeated two-point shadows, so they are instead reconstructed from their four incident blocks. The stabilizer of that block set has order $4 \cdot 48 = 192$, exactly the line stabilizer. Restriction to points is therefore faithful.

Both complete graph groups have order 88 704 000, and their derived subgroup is the specified image of G. Any geometry isomorphism carries the intrinsic graphs and their derived subgroups to one another. By faithfulness this conjugation lifts to the full geometries. It would therefore induce an equivalence of the specified actions, already excluded by the complete family comparison. This proves that all eighty-six representatives remain distinct as incidence geometries. The exact graph upper bound is a retained algorithmic premise, not a consequence of checking a few positive automorphisms.

The complete rank-one panel inventory has two faithful two-transitive actions, both of degree 176. Rank-one incidence is just a set, so they give one bare class with full group Sym(176). Both have actual truncation maps to higher-rank representatives. There are no other rank-one actions satisfying the source conditions.

For ranks at least two, inspect every proper type subset of every representative. There are 756 subsets. Intrinsic type sizes, pair-flag counts and chamber counts leave exactly fifteen comparisons with a listed target. Every surviving comparison has an actual whole-family automorphism and type map, hence an incidence isomorphism on cosets. No negative assertion relies on a failed

unbounded isomorphism search. The targets are two distinct rank-two classes and ten distinct rank-three classes; none have higher rank. Since the parent list is complete and any isomorphism transports its type subsets, this tests every possible truncation. Subtraction proves the second row of Theorem 1. The thirteen printed supplementary links are obtained by composing these maps with the source coordinate maps.

4 Residues, local groups and source coordinates

For each class, the type sizes and node orders are respectively $|G : H_i|$ and $|G_{I\setminus\{i\}} : B| - 1$. The kernel of H_i on its whole residue is

$$K_i = \bigcap_{j \neq i} \mathrm{Core}_{H_i}(H_i \cap H_j).$$

For a rank-two residue with group H, parabolics P, Q and $D = P \cap Q$, the neighbours of Px are exactly Qux, where u runs through a right transversal of D in P. This formula constructs the actual incidence graph without confusing equality of counts with equality of incidences.

All 437 base rank-two residues are acquired, with 187005 vertices and 601049 edges in total. A breadth-first search from one vertex in each part determines the type diameters because the action is transitive on each part. It also determines the girth here. For a shortest cycle, transitivity places one vertex at the chosen root. The opposite vertex and its two neighbours have the distances prescribed by the shortest cycle; otherwise a shorter path would create a shorter cycle. One of the two opposite edges is a nontree edge, and its two root distances plus one give the cycle length. Conversely every such nontree value is at least the girth. Thus the minimum is exact. The diagram triple is the first diameter, half the girth, and the second diameter.

The source coordinates give a bijection from all eighty-six diagram labels to the proved classes. Most are fixed by the type-size multiset. The tied diagrams $3.22, 3.23$ are distinguished by their proved full groups, and $5.2, 5.3$ by their complete bond patterns. The explicit source-to-model type permutations are retained. This is identification within a complete classification, not a claim that unavailable historical generators equal the current permutations. Thirteen Petersen residues and two affine-plane residues have literal adjacency-preserving bijections to the two-subsets model of Petersen and to $\mathrm{AG}(2, 4)$ over $\mathbb{F}_2[t]/(t^2 + t + 1)$.

The local structural certificates identify all 86 Borels and 312 residue kernels, followed by all 312 full parabolics with their displayed kernels and quotients. The elementary transport argument is worth making explicit. Enumerate the complete Cayley graph of a named finite model M using its generators. Proposed images y_m in the actual subgroup must satisfy $y_1 = 1$ and $y_{ms} = y_m y_s$ on every generator edge. Induction on positive words proves multiplicativity. Distinct images, containment of the actual generators and the acquired order prove an isomorphism onto the actual subgroup. A discovery tree alone would not suffice: every remaining edge is also checked.

For a quotient by a normal subgroup K, the corresponding equations hold modulo K. The distinct products ky_m prove the exact index and quotient map. Exact edge equations give a section; commuting with K additionally gives a direct product. To disprove splitting for a fixed quotient, every ordered tuple of possible generator lifts is tested. In the three relevant S_5 quotients all 4096 lift tuples fail, proving nonsplitting. Merely failing to find a complement would not prove this. Dot notation is treated as an extension, not as an assertion of nonsplitting. Named models and their construction proofs, including the separate $\mathrm{Aut}(A_6)$ identification, remain explicit dependencies of these transports.

5 Full automorphism and correlation groups

For a transitive type with base stabilizer P, an equivariant encoding of its objects is injective exactly when the stabilizer of its base encoding equals P. An object may first be encoded by its

point shadow, then by the sets of already identified incident objects. Induction in the recorded dependency order proves that an automorphism fixing the chosen point type fixes every object. This is the common faithfulness argument; repeated shadows are not silently treated as distinct.

Several intrinsic relations give upper bounds on the point action. Let the point type have size n, fix a base point x incident with the type-j base object, and let that object's point shadow be S_j. Let O_k be an orbit of G_x of size d_k. The number of type-j objects incident with a pair in the corresponding ordered-pair orbit is

$$m_{jk} = \frac{[G : P_j]e_{jk}}{nd_k}, \qquad e_{jk} = |S_j|\,|S_j \cap O_k|.$$

The first equality counts object–pair incidences; the second follows from the transitivity of P_j on S_j. These formulas count objects even if shadows repeat. Sorting the data $(|G : P_j|, m_{jk})$ over types gives a relation intrinsic under all permitted type permutations. The complete graph group often has the specified G as its derived subgroup. Normality of G, an abelian quotient and $G' = G$ imply this equality. An arbitrary correlation then normalizes G, by characteristicity on points and faithful restriction to the geometry.

A second bound uses the link of a point in a family $\mathcal{B}$ of distinct point subsets. If A preserves $\mathcal{B}$ transitively on the point set X, its stabilizer at x embeds in the permutation group of $X \setminus \{x\}$ preserving $\{B \setminus \{x\} : x \in B \in \mathcal{B}\}$. Hence $|A| \le |X||L_x|$. If this equals the known subgroup order, equality determines the full typed group. Still other cases use an induced incidence system on several intrinsically selected types. Complete coloured graph bounds and literal lifting of their generators settle the two set constructions separately.

Once the specified G is normal in the full correlation group, put $B = \bigcap_i P_i$ and let S, T be the ordered and setwise family stabilizers in $\mathrm{Aut}(G)$. A chamber-fixing automorphism of G induces the exact coset map $P_i g \mapsto \alpha(P_i)\alpha(g)$. Conversely move a correlation's image chamber back by G and use its induced automorphism of G. This proves

$$|\,\mathrm{Aut}(\Gamma)| = |G : B||S|, \qquad |\,\mathrm{Cor}(\Gamma)| = |G : B||T|.$$

Every type permutation is tested by the complete transporter procedure. The centralizer of G in the correlation group is the actual quotient $(T \cap G)/B$. Its intersection with the typed group is trivial in the relevant cases. A checked central subgroup of order two, or a checked elementary subgroup of order four, and equality of product orders then prove the displayed direct products. They are not inferred solely from doubled group orders.

Eight accepted applications cover 82 representatives. Their disjoint sets have sizes 24, 9, 6, 6, 11, 5, 15 and 6. Cases 14, 15 and the constructions 35, 36 cover the remaining four. The full source table in the ledger contains all eighty-six names and their exact proof references. The following printed fields change; put $A = \mathrm{Aut}(\mathrm{HS})$.

Diagram	Field	Printed	Proved
3.20	Aut	G	A
3.20	Cor	A	$A \times C_2$
3.21	Aut	G	A
3.21	Cor	$G \times C_2$	$A \times C_2$
3.26	Aut	G	A
3.26	Cor	A	$A \times C_2$
4.22	Cor	$G \times C_2$	G
4.41	Cor	$A \times C_2$	A

The two blank fields of 3.10 are filled by A and $A \times C_2$. They are completed metadata, not two false printed groups.

6 The two Hoffman–Singleton constructions

In the literal model take 176 points, 1100 conics, each an eight-set, and 176 quadrics, each a fifty-set. The complete saved orbits verify the six Higman properties used in the article. Each point lies on fifty quadrics, and each conic on eight. Every point pair lies on two conics; the quadrics through that pair are exactly the union of those through the two conics, with precisely two quadrics containing both conics. Dually each quadric pair contains two conics, its point intersection is their union, and the conics meet in two points.

The sixth property is checked with both of its universal clauses. For a base conic C, form the 8 by 8 relation between its points and its incident quadrics: all conics meeting C in two points including the chosen point must lie in the chosen quadric; all conics lying in two quadrics through C, one the chosen quadric, must contain that point. Exactly one pair in each row and column satisfies both clauses. Transitivity on conics transports this matching to every conic. Literal point and quadric actions are doubly transitive, and the point stabilizer has quadric orbits of lengths $50, 126$, proving the additional transitivity assertions.

Let the lines be all $\binom{176}{2} = 15400$ point pairs. The first geometry has points, lines and conics, with containment incidence. The second has points, lines and quadrics. Point incidence is containment; a line meets a quadric exactly when the quadric contains both conics through that line. For each construction, the map from an actual coset $H_i g$ to the translate of its base subset is checked as a bijection. For every ordered pair of types, the subgroup orbit of a base object equals its entire geometric neighbour set. Equivariance proves that these maps preserve incidence and nonincidence everywhere. They identify the constructions with representatives $35, 36$, respectively, and with supplement diagrams $3.2, 3.1$.

Every line has two points and two planes; all four flags in its residue occur. Every incident point–plane pair has seven common lines. Thus all partial flags extend, and both geometries have 61600 chambers with Borel S_6. In the first geometry a point residue is a graph on fifty conics with 175 distinct edges, valency seven, diameter two and girth five; its conic residue is literally K_8. In the second, both point and quadric residues have the same fifty-vertex graph parameters. The classical uniqueness of the Hoffman–Singleton graph identifies these graphs. Their full group $U_3(5) : 2$ is retained as a named classical input; faithful actions of the actual parabolics attain its order. The line residues are digons. The line stabilizers have a literal S_6 kernel and two commuting involutions giving a disjoint complement of order four, proving $S_6 : 2^2$.

For the first construction the coloured point–conic graph has full group of order $|G|$; all its generators lift to the uniquely determined lines. Distinct type sizes exclude additional type permutations, so $\mathrm{Aut} = \mathrm{Cor} = G$. For the second, the marked point–quadric graph has full group of order $2|G|$. Its generators lift to lines using the verified bijection from point pairs to quadric pairs. The type kernel is G, and its centralizer is trivial, so conjugation identifies the full group with $\mathrm{Aut}(G)$. A saved type exchange supplies the duality. These constructions give the partial answer to the cited extension problem claimed by the original article; they are not newly solved open problems of this project.

7 Ancillary source statements and corrections

The 18 conjugacy notes involve 24 actual pairs of parabolics inside G: eleven positive transports and thirteen complete negative decisions. In 4.41 both pairs on the same level are conjugate and all four cross pairs are not. Abstract subgroup names alone do not decide these relations. The three stronger RPRI annotations, at $3.1, 3.2, 4.1$, require all inclusions $G_{J\cup\{i\}} < G_J$ to be maximal, not just one type per residue. All 56 inclusions are transported to actual rows in the complete maximal-inclusion catalogue. Finally each parenthetic source number is joined individually to its inner/type fusion fibre: 42 ones and 44 twos sum to 130. They are not additional incidence classes.

For doubling, the precise operation replaces a point type X and a type of distinct two-point

shadows by two copies X^+, X^-. An old edge joins the corresponding opposite copies; incidences with remaining types are inherited from the endpoints. When an edge is incident with another object exactly when both endpoints are, an old chamber gives a new chamber and conversely. The finite tests prove this condition, shadow injectivity and the whole-family target maps for

$$3.2 \to 3.3, \quad 3.4 \to 3.5, \quad 3.8 \to 3.9, \quad 4.8 \to 4.10, \quad 4.14 \to 4.19.$$

For $3.12 \to 3.15$ and $3.13 \to 3.16$, the only size-compatible choice has ten-point shadows. For $3.20 \to 3.21$, no choice can produce the new type size 30800. These are obstructions to the specified operation. The target geometries still exist, and no unspecified repair is claimed.

Six supplementary diagrams require numerical amendments. At 2.5 the relevant node order is 15, not 11; at 2.6 the type size is 44352, not 44320. The indicated triple in 3.15 is $(5, 4, 6)$ rather than $(6, 4, 5)$, and the top node order in 3.30 is 3, not 1. Diagram 4.22 has a digon at the lower bond and an omitted single diagonal bond between its 352-object types. In 4.28 the two indicated bonds have triples $(7, 5, 8)$ and $(4, 3, 3)$, the latter with the complete-graph incidence interpretation. Every correction is checked against all compatible type permutations in the complete class list.

The structural ledger retains eleven corrected occurrences of Borels or kernels and twenty-six parabolic occurrences. Five of the latter repeat the already corrected boxed kernels and are linked to those same occurrences; they are not counted again. The remaining parabolic clauses include three false splitting colons, ten affine quotients incorrectly printed as direct products, and the explicit quotient or full-group replacements listed in the ledger. Actual group maps prove the replacements; order, centre, element-order and complete-section arguments disprove the incompatible printed alternatives. A differently parenthesized but valid intermediate product in 4.12 is retained as correct.

Our own documentary mistakes are recorded separately. An old Table 1 transcription gave 153483 subgroups for J_1; the source actually prints 158485, agreeing with the proof. The intermediate product in 4.12 was once incorrectly read as a central direct factor of the full group. An auxiliary Janko transcription silently replaced a printed 17742 by the correct 17442; the source error and that transcription mistake are both retained. None of these is folded into a grand source-error count.

8 The comparison with the other sporadic groups

The article expressly quotes earlier classifications for the seven non-HS rows of Table 1. For a complete subgroup catalogue, the number of subgroups is $\sum_H [G : N_G(H)]$. In a complete table of marks its class lengths are $m_{H,1}/m_{H,H}$, because the two marks are $[G : H]$ and $[N_G(H) : H]$. Exact acquired rows give the following table. The rank columns remove only truncations of eligible higher-rank geometries in the same locally two-transitive class.

G	$\lvert G\rvert$	Classes	Subgroups	2	3	4	5	6
M_{11}	7920	39	8651	0	9	11	0	0
M_{12}	95040	147	214871	5	10	17	1	0
J_1	175560	40	158485	11	1	2	0	0
M_{22}	443520	156	941627	1	20	12	9	0
J_2	604800	146	1104344	5	16	3	0	0
M_{23}	10200960	204	17318406	0	3	16	10	0
HS	44352000	589	149985646	4	20	41	8	1
J_3	50232960	137	71564248	5	19	0	0	0

For the first six groups, the retained geometric equivalence is under $\mathrm{Aut}(G)$ and type permutations. No HS bare-isomorphism theorem is transferred to them without proof. The M_{12} rank-three value is 10 in that precise convention, compared with the printed 11. The independent HS argument above does prove its bare quotient.

The J_3 comparison has its own explicit boundary. Twenty-one acquired rank-three families have nineteen bare classes: intrinsic flag counts, two common-point-graph separations and IP/WIP separate them, while two actual whole-family maps identify the only repeated pairs. Its seven rank-two prescriptions are eligible and have distinct object-size pairs. All sixty-three type-pair profiles from the rank-three roster match exactly the two literal truncations already identified. This proves seven minus two equals five within that supplied population.

To interpret the last row globally, we retain from the article's reference [9] the earlier Janko classification's coverage and its absence of eligible geometries of rank at least four. The printed number seventeen is not imported as a premise: it counts diagram labels rather than the nineteen proved supplied classes. Reference [9] is historical L097, outside the fixed active 105 mathematical originals because its full article is unavailable; its acquired thirteen-page author supplement is auxiliary. No independent global Janko census or L097 recovery is claimed. The other auxiliary Janko metrics, names and full groups are not new assertions of L091 and are not additional conditions for its comparison-table disposition.

9 Attribution, premises and the proof volume

The following boundaries apply throughout.

1. The named faithful HS model, its complete subgroup catalogue, the complete maximal-inclusion lists, and the acquired full order and action of Aut(HS) are retained finite-group inputs. Their exact sources, generators, completed executions and review records are bound. The panel and maximal-deletion theory is reproduced here and in the common proof chapters; it is not replaced by trusting the target classification's printed totals.

2. Named small groups and extension models use their acquired construction or classification proofs. In particular the $\mathrm{Aut}(A_6)$ addendum is required in addition to the general parabolic transport certificates. Generic extension notation without its actual action is not treated as a unique group identifier. The addendum retains $A_6 \cong \mathrm{PSL}_2(9)$ and the full automorphism theorem identifying $\mathrm{Aut}(\mathrm{PSL}_2(9))$ with $\mathrm{P\Gamma L}_2(9)$. Conjugating automorphisms by the first isomorphism gives the required identification. The saved degree-ten semilinear model has exactly the same ordered generators as model 7003 in the earlier structural proof and model 319 in the HS library. The literal transports therefore land in that particular named group. Neither order 1440 nor the string $A_6 : 2 : 2$ supplies this identification by itself.

3. GAP's exact finite-group, subgroup, coset and transporter algorithms are software premises. Complete nauty or Traces upper bounds are also software premises. Literal positive generators, adjacency checks and small independent controls supplement these bounds; they do not alone prove completeness. No Lean verification is claimed by this intake.

4. The classical identification of the fifty-vertex Moore graph and its group $U_3(5) : 2$, and the historical Higman–Sims identification, retain their attribution to the cited work. The source's partial answer to the Hoffman–Singleton extension problem remains partial and is not counted as a new conjecture solved here.

5. The supplement attributes 3.4 to Ivanov–Shpectorov, 4.1 to Buekenhout's 1979 work, and 6.1 to the separate rank-six paper cited as reference [13] by the article. The present exact models recover their displayed occurrences but do not silently claim to reproduce those external constructions or complete another original.

The old mathematical results remain the work of the original authors. The present contributions are the common proof organization, the exact finite reproving and transports, the intrinsic incidence arguments that separate action equivalence from geometry isomorphism, and the explicit source corrections. These elementary arguments are not advertised as historically new solely because they appear in this project.

The complete proof volume preserves the accepted chapters unchanged. Their earlier local scope statements are historical; only the present occurrence ledger declares which later proof discharges each remaining clause. The acquisition manifest binds every included chapter and its full evidence chain. The final verifier checks that the source classes, local slots, group cases, annotations and correction links are complete and nonduplicated, then recompiles this synthesis. It does not repeat already accepted subgroup or graph searches. Original-level admission still requires a distinct review of this assembly. No compression factor or additional conjecture credit is assigned by joining these components.

10 Complete source class and group index

Here $G = \mathrm{HS}$ and $A = \mathrm{Aut}(\mathrm{HS})$. The numeric identifier fixes the literal family; the source-to-model type map and every local structural occurrence are in the accompanying ledger. An asterisk marks a corrected full-group field; a dagger marks the two completed blank fields.

Diagram	Page	Family	Multiplicity	Aut	Cor
2.1	3	6	1	A	A
2.2	3	5	1	A	A
2.3	3	4	1	A	A
2.4	3	3	1	G	A
2.5	3	2	1	A	A
2.6	3	1	1	A	A
3.1	4	36	1	G	A
3.2	4	35	2	G	G
3.3	4	34	2	G	$G \times C_2$
3.4	4	32	1	A	A
3.5	4	33	1	A	$A \times C_2$
3.6	5	31	1	A	A
3.7	5	29	1	A	A
3.8	5	30	1	A	A
3.9	5	28	1	A	$A \times C_2$
3.10	5	27	1	$A^\dagger$	$A \times C_2{}^\dagger$
3.11	6	26	1	A	$A \times C_2$
3.12	6	24	2	G	G
3.13	6	21	2	G	G
3.14	6	23	2	G	G
3.15	6	25	2	G	$G \times C_2$
3.16	7	22	2	G	$G \times C_2$
3.17	7	19	2	G	G
3.18	7	20	2	G	G
3.19	7	18	1	A	A
3.20	7	17	1	A^*	$A \times C_2{}^*$
3.21	8	16	1	A^*	$A \times C_2{}^*$
3.22	8	14	2	G	G
3.23	8	15	1	A	A
3.24	8	13	2	G	G
3.25	8	11	1	A	A
3.26	8	12	1	A^*	$A \times C_2{}^*$
3.27	9	10	1	G	A
3.28	9	9	2	G	G
3.29	9	8	2	G	G
3.30	9	7	2	G	$G \times C_2$
4.1	10	77	1	A	$A \times C_2$
4.2	10	76	1	A	A
4.3	10	74	1	A	A
4.4	10	75	1	A	A

Diagram	Page	Family	Multiplicity	Aut	Cor
4.5	11	73	1	A	$A \times C_2$
4.6	11	72	1	A	$A \times C_2$
4.7	11	71	1	A	$A \times C_2$
4.8	11	70	2	G	G
4.9	11	68	2	G	G
4.10	12	69	2	G	$G \times C_2$
4.11	12	67	1	G	A
4.12	12	66	2	G	G
4.13	12	60	1	A	$A \times C_2$
4.14	12	54	2	G	G
4.15	13	55	2	G	$G \times C_2$
4.16	13	59	2	G	$G \times C_2$
4.17	13	58	2	G	G
4.18	14	56	2	G	$G \times C_2$
4.19	14	57	2	G	$G \times C_2$
4.20	14	65	2	G	G
4.21	14	64	2	G	G
4.22	15	63	2	G	G^*
4.23	15	62	2	G	G
4.24	15	61	2	G	G
4.25	15	53	2	G	$G \times C_2$
4.26	16	52	2	G	$G \times C_2^2$
4.27	16	51	2	G	G
4.28	16	38	2	G	G
4.29	16	39	2	G	G
4.30	17	40	1	A	A
4.31	17	37	2	G	$G \times C_2$
4.32	17	49	1	A	A
4.33	17	50	1	A	A
4.34	18	48	2	G	G
4.35	18	47	2	G	G
4.36	18	43	2	G	G
4.37	18	44	2	G	G
4.38	19	41	2	G	G
4.39	19	45	1	G	A
4.40	19	42	1	G	A
4.41	20	46	1	G	A^*
5.1	20	85	1	A	$A \times C_2$
5.2	20	83	2	G	$G \times C_2$
5.3	21	84	2	G	$G \times C_2$
5.4	21	81	1	A	$A \times C_2$
5.5	21	82	1	A	$A \times C_2$
5.6	21	78	1	G	A
5.7	22	80	2	G	G
5.8	22	79	2	G	G
6.1	22	86	1	A	$A \times C_2$

11 Index to the complete proof volume

The companion volume appends the following accepted chapters without changing their mathematical text. The exact author and review records, including role disclosures, are in the acquisition manifest.

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13	Groups from one hundred points
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19	Fifteen groups from larger point actions
20	The last six geometry groups
21	The two Hoffman–Singleton constructions
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24	The first seven comparison rows
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A Accepted proof chapters

The appended pages retain their original local pagination and scoped statements. The source ledger above records the later closures; no earlier partial scope is silently promoted.

Stabilizers, extensions and the enumeration of finite geometries

Human Compression project

Abstract

A classification by group actions requires both an equivalence relation and an admissibility test. We give a common way to extend a partial configuration, impose a support condition, and count its orbits using its actual stabilizer. We also prove recognition and exhaustive augmentation for coset geometries with an arbitrary chamber stabilizer. The applications concern commuting involutions, Suzuki polyhedra and the residually weakly primitive geometries of the Hall–Janko group. Their local group structure and exact computations remain explicit inputs.

1 The information a classification must retain

Two commuting involutions do not yet specify a regular polyhedron. A third involution must generate the prescribed group with them and satisfy all subgroup intersections. Likewise, panel subgroups do not specify a flag-transitive geometry until incident cosets have common representatives. The method developed here keeps these tests separate from the orbit reduction.

After fixing a partial configuration, its stabilizer acts on the remaining choices. Theorem UD.AdmissibleExtensions counts precisely the admissible choices, including a prescribed field of definition when present. It combines the elementary extension-orbit correspondence with the classical fixed-point count and Möbius inversion. Its role is to expose which hypotheses each application must prove; no priority is claimed for these classical principles. The finite-action theorem also records when a permutation image is faithful, so that an abstract automorphism group is not confused with a normalizer in one chosen action.

For coset geometries, the common paper contains the complete recognition and augmentation proofs. The Hall–Janko application supplies the subgroup data, the actual stabilizer actions and the finite outputs. For Suzuki groups, normalization and a subfield-position argument identify the support condition with generation. For symmetric and alternating groups, the transitive four-group actions supply the multiplicities and parity conditions. These local arguments are part of their applications rather than assumptions hidden inside the common theorem.

All groups in the counting statements are finite. The extension-orbit correspondence and the elementary coset realization are stated without that restriction. A geometry has a finite type set; incidence is reflexive and symmetric and two elements of the same type are incident only when equal. A flag is a pairwise incident set, a chamber has every type, and every flag must extend to a chamber. Firmness means that a flag missing one type extends in at least two ways. Residual connectedness concerns the incidence graph of every residue of rank at least two. The group action is type preserving; flag transitivity means transitivity on flags of every prescribed type.

A C-group marking is a generating family of distinct nonidentity involutions $(r_i)_{i\in I}$ such that, with $G_J = \langle r_i : i \in J\rangle$ and $G_\varnothing = 1$,

$$G_J \cap G_K = G_{J\cap K} \qquad (J, K \subseteq I).$$

For $I = \{0, \dots, d-1\}$ it is a string marking if $r_i r_j = r_j r_i$ whenever $|i - j| \geq 2$. The applications retain the standard correspondence between string C-group markings and regular abstract polytopes. Reversing the ordered markers corresponds to duality. A rank-three marking is therefore a regular polyhedron only after generation and all intersections have been established; endpoint orders alone do not give this conclusion.

2 Finite actions and extensions

Theorem UD.Orbits (orbit data and normalizers). Let a finite group A act on a finite set Ω, and choose one representative H of each conjugacy class of subgroups that occurs as a point stabilizer. Let m_H be the number of orbits isomorphic to A/H, with $m_H = 0$ for all omitted classes. Then the numbers m_H determine the A-set up to equivariant isomorphism, and

$$\ker(A \curvearrowright \Omega) = \bigcap_{m_H>0} \operatorname{core}_A(H). \tag{5.3}$$

Its equivariant permutation group is abstractly

$$C_{\mathrm{Sym}(\Omega)}(A) \cong \prod_H \big((N_A(H)/H) \wr \mathrm{Sym}(m_H)\big). \tag{5.4}$$

If the action $\rho : A \hookrightarrow \mathrm{Sym}(\Omega)$ is faithful, conjugation gives an exact sequence

$$1 \longrightarrow C_{\mathrm{Sym}(\Omega)}(\rho(A)) \longrightarrow N_{\mathrm{Sym}(\Omega)}(\rho(A)) \longrightarrow \{\alpha \in \mathrm{Aut}(A) : m_{\alpha(H)} = m_H \text{ for all } H\} \longrightarrow 1. \tag{5.5}$$

No splitting of this sequence is asserted.

Proof. Every orbit is equivariantly A/H, where H is the stabilizer of a selected point, and two such transitive sets are isomorphic exactly when the stabilizers are conjugate. This proves the classification by multiplicities and the kernel formula.

An equivariant permutation preserves the isomorphism type of each orbit. For one orbit A/H, an equivariant map is determined by the image of H. That image nH must be fixed by H, and bijectivity makes its stabilizer equal to H; hence $n \in N_A(H)$. Conversely each such n gives a permutation $aH \mapsto anH$. Composition corresponds to multiplication in the opposite order, so inversion identifies this permutation group with $N_A(H)/H$. For m_H copies, one may independently choose such a map in every copy and permute the copies, yielding the wreath product. Different isomorphism types are independent, proving (5.4).

Faithfulness makes conjugation by a normalizing permutation τ induce a unique automorphism α of A, with

$$\tau\rho(a)\tau^{-1} = \rho(\alpha(a)).$$

If a point has stabilizer H, its image under τ has stabilizer $\alpha(H)$; therefore the multiplicities must be invariant. The kernel of this conjugation map is the centralizer. Conversely, suppose α preserves the multiplicities. Pair each orbit with a target orbit whose stabilizer class is $\alpha(H)$, and choose base points x, y with actual stabilizers $H, \alpha(H)$, adjusting within the target orbit if needed. Define $\tau(ax) = \alpha(a)y$. The stabilizer equality makes this well-defined and bijective on that pair of orbits, and the paired definitions combine to a permutation of Ω. It satisfies the displayed intertwining equation, proving surjectivity and exactness. □

Proposition UD.ExtensionOrbits (classification above a retained base). Let a group B act on sets Z, E, and let $p : Z \to E$ be equivariant. For one representative e of each orbit in $B\backslash E$, there is a natural bijection

$$B\backslash Z \;\cong\; \coprod_{e \in B\backslash E} B_e\backslash p^{-1}(e). \tag{5.6}$$

Proof. Translate any $z \in Z$ so that $p(z)$ is the selected representative of its orbit. If two translated points over that representative are B-equivalent, an element taking one to the other fixes their image e, so belongs to B_e. Conversely a B_e-equivalence is already a B-equivalence. These two observations prove surjectivity and injectivity of (5.6). □

3 Admissible extensions with prescribed support

Let a finite group A act on finite sets Z and E, and let $p : Z \to E$ be equivariant. Let $R \subseteq Z$ be invariant; it is the set passing the application's admissibility tests. Suppose a finite poset L with greatest element 1 has an A-action by order automorphisms and an equivariant map $s : R \to L$. Write $R_l = \{z \in R : s(z) \le l\}$. The individual sets R_l need not be invariant. Choose one e in each A-orbit in $p(R)$, and put $R(e) = R \cap p^{-1}(e)$.

Theorem UD.AdmissibleExtensions. The equivalence classes of admissible extensions of exact support 1 are in bijection with the disjoint union of the A_e-orbits on $\{z \in R(e) : s(z) = 1\}$. Their number is

$$\sum_{e \in A\backslash p(R)} \frac{1}{|A_e|} \sum_{a \in A_e} \sum_{l \in L} \mu(l,1) |\operatorname{Fix}_{R(e)}(a) \cap R_l|,$$

where μ is the Möbius function of L.

Proof. The greatest element is fixed by every order automorphism, so the exact support condition is invariant. Apply UD.ExtensionOrbits to the restriction of p to that subset. Its fiber equivalence is exactly the A_e-action: an element carrying one extension over e to another necessarily fixes e. An empty exact-support fiber contributes zero, so representatives may be taken from all of $p(R)$.

For each $z \in R$, the defining inverse identity of the incidence-algebra Möbius function gives

$$\sum_{s(z) \le l \le 1} \mu(l,1) = \begin{cases} 1, & s(z) = 1, \\ 0, & s(z) < 1. \end{cases}$$

Sum over the points of $R(e)$ fixed by a particular $a \in A_e$. This gives the inner sum in the formula. Finally, for a finite group H acting on a finite set Y, count the pairs (h, y) with $hy = y$. Their number is both $\sum_h |\operatorname{Fix}_Y(h)|$ and $\sum_y |H_y|$. Each orbit contributes $|H|$ to the second sum by the orbit–stabilizer identity. Division by $|H|$ gives the number of orbits. Use this identity with $H = A_e$ and the exact-support fiber, and sum over the representatives e. □

Corollary UD.ExactSupportOrbits. If A acts on a finite set X with an equivariant support map to L, the number of exact-top-support orbits is

$$\frac{1}{|A|} \sum_{a \in A} \sum_{l \in L} \mu(l,1) |\operatorname{Fix}_X(a) \cap \{x : s(x) \le l\}|.$$

Proof. Take E to be a one-point set and $R = Z = X$. □

Taking a one-element support poset instead recovers counting in ordinary extension fibers. For an involution D on a finite set Y, it gives $(|Y| + |\operatorname{Fix}_Y(D)|)/2$ orbits. A division by two is therefore justified without a correction term only after freeness has been proved.

Lemma UD.OddOrbits. Let a finite group A of odd order act on X. Let D be a fixed-point-free involution of X commuting with this action. Then D has no fixed orbit in $A\backslash X$.

Proof. If $Dx = ax$, commutation and $D^2 = 1$ give $a^2 x = x$. The order of a is odd, so a is a power of a^2 and fixes x. Then $Dx = x$, a contradiction. □

For the divisor poset of an integer N, support often means the least field $\mathbb{F}_{2^d}$ containing all coordinates, where $d \mid N$. The application must establish that its subfield sets are exactly the support sublevel sets. If $A(d)$ counts data defined over that field, the number of data of exact degree N is

$$E(N) = \sum_{d \mid N} \mu(N/d) A(d).$$

If Frobenius acts freely on them with orbit length N, division by N counts its orbits. For odd N, a commuting fixed-point-free involution then permits another division by two, by UD.OddOrbits. These are conditional consequences of the common theorem, not substitutes for the generation and stabilizer calculations in the applications.

4 Cosets and flags

For subgroups $P_i \le G$, typed coset Helly means that every pairwise intersecting family of right cosets $P_i g_i$, with distinct indices, has a common element. This property is separate from IP. It supplies a common representative for an incident flag, whereas IP determines the stabilizer of a flag already represented in this way.

Lemma UD.Helly (anchored criterion). A finite indexed subgroup family $(P_i)_{i\in I}$ has typed coset Helly if and only if the following condition holds. For every $i \in I$, every $J \subseteq I \setminus \{i\}$ with $|J| \ge 2$, and any selected $j_0 \in J$, each $g \in P_{j_0}$ satisfying $g \in P_i P_j$ for all $j \in J$ admits $k \in \bigcap_{j\in J} P_j$ with $P_i g = P_i k$.

Proof. The cosets $P_i g$ and P_j intersect precisely when $g \in P_i P_j$. Thus Helly gives a common element k, proving necessity. Conversely, induct on the number of cosets in a pairwise intersecting family. A family of at most two already has a common element. For a larger family, the induction hypothesis gives a common element of all but one coset. Right multiplication by its inverse turns those cosets into the base cosets P_j. The remaining coset meets P_{j_0}, so it has a representative $g \in P_{j_0}$. Its intersections with the other base cosets are precisely the product conditions in the statement. The criterion supplies k, and undoing the right translation gives a common element of the original family. □

5 Panel systems with a nontrivial chamber stabilizer

Lemma UD.CosetRealization. Let a group G act type-preservingly and flag-transitively on an incidence geometry of finite rank, and let $(x_i)_{i\in I}$ be a base chamber. Write $H_i = \mathrm{Stab}_G(x_i)$ and use a right action. Then

$$H_i g \longmapsto x_i^g$$

is a type-preserving G-equivariant isomorphism from the subgroup coset system to the given geometry.

Proof. Transitivity on one-object flags makes the map onto each type. It is well defined and injective because $x_i^g = x_i^h$ holds exactly when $gh^{-1} \in H_i$, which is equivalent to $H_i g = H_i h$. If two cosets share a representative k, their images are x_i^k, x_j^k and are incident, as the base objects are incident. Conversely, if two images are incident, transitivity on flags of their two types supplies a k carrying the base pair to them. The injective object correspondence then writes both cosets with that common representative. The map therefore preserves and reflects incidence. □

Let G be finite, let I be a finite type set, and let each panel subgroup P_i contain B. Put $K_\emptyset = B$ and $K_S = \langle P_i : i \in S\rangle$ for nonempty S, with $K_I = G$. This marking by subgroups is different from a marking by individual involutions.

Theorem UD.PanelRecognition. Suppose

$$K_S \cap K_T = K_{S\cap T} \quad (S, T \subseteq I), \qquad [P_i : B] \ge 2, \tag{E4}$$

and suppose the maximal parabolics $H_i = K_{I\setminus\{i\}}$ have typed coset Helly. Then their coset incidence system is a firm flag-transitive residually connected geometry with chamber stabilizer B. Conversely every finite-rank firm flag-transitive residually connected coset geometry is obtained this way by taking the panel stabilizers.

Proof. Equation (E4) identifies the stabilizer of every partial base flag with the corresponding K_S. UD.Helly supplies a common group element in the cosets of every flag, and hence flag transitivity and identification of the residues with the corresponding coset systems in K_S. Each rank-one residue has $[P_i : B]$ elements, giving firmness. A rank-two residue is connected because its two panel subgroups generate $K_{\{i,j\}}$. For larger rank, all panel subgroups occur in proper deletions, so the residue group is generated by its maximal parabolics. The usual coset-path argument proves connectedness: paths multiply successive elements of those subgroups, and conversely their products give paths.

Conversely let K_S stabilize the complementary base flag. Stabilizer intersections give (E4). In a connected residue, the subgroup generated by its maximal parabolics has exactly the orbit reachable by

coset paths, so it is the whole residue group. Induction on $|S|$ reconstructs K_S from the panels, starting with $K_\emptyset = B$. Flag transitivity gives typed Helly by moving a flag to the base flag. □

The source's remaining axioms are separate, exact local tests. RWPRI means that for every nonempty S there is an $i \in S$ with $K_{S\setminus\{i\}}$ maximal proper in K_S. Local two-transitivity means that P_i has two orbits on ordered pairs in $B\backslash P_i$; equivalently B has two orbits on that coset set. The source's geometric $(\mathrm{IP})_2$ means that every rank-two residue is a partial linear space or a generalized digon. It is not the subgroup identity (E4).

For the last test let P, Q be two panels, $P \cap Q = B$ and $K = \langle P, Q\rangle$. The coset graph is a digon exactly when $|P||Q| = |K||B|$. Otherwise local two-transitivity makes a single common-neighbour count sufficient. Choose $x \in Q \setminus B$. Neighbours of P are the Q-cosets represented by a right transversal of B in P; neighbours of Px are their translates by x. The stabilizer Q is two-transitive on its incident P-type points, so all distinct points having a common Q-type neighbour have the same number of common neighbours. This number must be one. Pairs with no common neighbour already satisfy partial linearity. This argument also applies when a local action has a nontrivial kernel.

5.1 Complete augmentation through maximal residues

Lemma UD.Transport. Suppose representatives of all ambient G-conjugacy classes of subgroups K and all K-conjugacy classes of maximal subgroups $H_0 < K$ are given. For a specified H, choose any transporter c with $H_0^c = H$ when one exists. All maximal overgroups of H in that inclusion class are precisely $(K^c)^n$ with $n \in N_G(H)$.

Proof. Conjugate any maximal inclusion $H < L$ to a representative K of the class of L. Its image of H is K-conjugate to one of the listed H_0. Comparing the resulting transporter with c leaves an element normalizing H. Conversely each displayed conjugate contains H maximally. Taking the union over all inclusion classes gives every maximal overgroup. □

Let $\mathcal{P}(B)$ be the finite set of overgroups P of B such that B is maximal proper in P and P acts two-transitively on $B\backslash P$. UD.Transport computes this entire set by retaining only the relevant local inclusions. The normalizer $N_G(B)$ acts on $\mathcal{P}(B)$, and its permutation image has exactly the same configuration orbits. We retain the actual subgroups behind the permutation labels, so kernel reduction does not change any geometric test.

Theorem UD.MaximalAugmentation. Start with all rank-one panels modulo $N_G(B)$. For every accepted panel tuple, let H be its generated join. For every maximal overgroup K of H, try every $P \in \mathcal{P}(B)$ with $P \le K$ and $P \not\subseteq H$, modulo the pointwise stabilizer of the old panel tuple. Apply UD.PanelRecognition and all the local source tests, then remove duplicates under $N_G(B)$ and type permutations. Iterating until the next level is empty gives every target geometry with Borel B and acting group a subgroup of G.

Proof. For a target rank-r geometry, RWPRI supplies at least one maximal parabolic H in its acting group K. Deleting its complementary panel gives a rank-$(r-1)$ residue with the same Borel and all inherited source axioms. Induction puts its panel tuple in a previously enumerated orbit. Its remaining panel lies inside K and outside H, and UD.Transport includes K among the maximal overgroups to be tried. UD.ExtensionOrbits shows that taking pointwise-stabilizer representatives loses no extension; a smaller stabilizer than the full setwise one merely permits duplicates. Final orbit tests remove those duplicates. Conversely every accepted tuple satisfies exactly the recognition and local conditions.

By (E4) and proper panels, adjoining a new type strictly enlarges the generated join: if P_i lay in K_S for i outside S, then $P_i \cap K_S = B$ would force $P_i = B$. Hence every tuple gives a chain of strict subgroup inclusions from B. Its rank is at most $\Omega([G : B])$, the number of prime factors with multiplicity. Thus a run through the next level must reach an empty level. The certificate requires that emptiness, not merely a chosen numeric cap. Finally take B over all G-conjugacy classes of subgroups. □

Attribution and use

The orbit classification, centralizer description, fixed-point count and Möbius inversion are classical. The common extension correspondence and the arbitrary-chamber-stabilizer recognition and augmentation proofs are reused from the previously reviewed project proof bank (U.Orbits, U.ExtensionOrbits and the enumeration manuscript, Sections 2 and 5). The combined counting interface removes separate invocations of those principles; it does not establish new historical classification results. Coset geometry and the equivalence between string C-groups and abstract regular polytopes retain their established meanings; the latter correspondence is an external input in the polyhedron applications.

The full common paper is charged once within this topic. Its overlap with other topic theories does not reduce that charge. Complete local proofs, required programs and the separately identified diagnostics accompany the applications. No numerical compression factor is assigned by this edition.

A bound from the group generated by all panels

Human Compression project

Fix a finite group G and a subgroup B. Suppose that an acquired complete transport procedure supplies the actual list $\mathcal{P}_B$ of admissible panel subgroups above this fixed B. We consider only families of these panels that generate G. The completeness of the panel list, including its local action conditions, is an input to the following argument, not a conclusion of it.

Lemma 1. *Put $J_B = \langle B, P \mid P \in \mathcal{P}_B \rangle$. If $J_B < G$, no family under consideration exists at B. If every $P \in \mathcal{P}_B$ has index two over B, then $J_B \leq N_G(B)$, so a proper normalizer suffices for the same conclusion.*

Proof. Every selected panel lies in J_B. The group generated by any selected family therefore lies in J_B, and cannot equal G when $J_B < G$. In the second assertion, a subgroup of index two is normal. Thus each P normalizes B, and the same holds for the group they generate together with B. □

For the connected coset geometries in the acquired primitive augmentation theorem, generation of G by the panel subgroups is part of the established correspondence. The list $\mathcal{P}_B$ is obtained by the already proved transport of complete maximal inclusions; for the Higman–Sims source it is restricted to the inclusions with a two-transitive local action. The bound applies to this exact list. It makes no assertion that proper generated families are absent, and supplies no classification of them.

The implementation computes the same complete actual panel list as the accepted enumeration, forms J_B, and tests properness. An exclusion saves explicit generating permutations for B, every panel, and J_B. It has its own terminal status. It is not an empty final level of the old enumeration. If $J_B = G$, the wrapper invokes the unchanged old procedure. This may repeat the inexpensive construction of its panel list; it changes no recognition, equivalence or pruning rule inside that procedure.

For example, the retained Higman–Sims case numbered 96 has $|B| = 16$ and 19 admissible panels, all of order 32. Their generated group has order 512, strictly less than $|G| = 44\,352\,000$. The completed lower levels of an interrupted enumeration remain valid, but are unnecessary for this exclusion. The unfinished rank is not credited as exhausted. The exact subgroup classification and finite-group software premises used to obtain the panel list remain explicitly acquired.

This is an elementary bound used to avoid unnecessary enumeration. No novelty, open-conjecture, whole-source or compression claim is made by the argument alone.

Generation by panels compatible with a partial family

Human Compression project

Let $\mathcal{P}_B$ be the acquired complete panel set for a fixed $B \leq G$. Write $P \sim Q$ when the acquired rank-two test accepts the distinct panels P, Q. Every family under consideration has mutually compatible panels. This is a necessary condition; no converse is used. For a selected family $T \subseteq \mathcal{P}_B$, put

$$C(T) = T \cup \{P \in \mathcal{P}_B \setminus T : P \sim Q \text{ for every } Q \in T\}, \qquad J(T) = \langle B, P \mid P \in C(T) \rangle.$$

Lemma 1. *If $J(T) < G$, no admissible family containing T generates G. The exclusion is invariant under conjugation by $N_G(B)$ and under reordering the selected panels.*

Proof. Any additional panel in an admissible family must be compatible with each member of T, and hence lies in $C(T)$. The whole family therefore generates a subgroup of $J(T)$. Properness proves the first assertion. The acquired rank-two conditions are properties of actual subgroup intersections, joins, maximality and coset incidence; conjugation preserves each of them. Thus conjugation sends $C(T)$ onto the compatible set of the conjugate family, and sends $J(T)$ to its conjugate. Reordering T changes neither set. □

Consequently this exclusion can be inserted into the acquired maximal augmentation search for families generating G. For any such final family F, the acquired augmentation theorem supplies a chain of admissible prefixes leading to a conjugate and reordering of F. For each prefix T on that chain, $F \subseteq C(T)$ and therefore $J(T) = G$. No required chain is removed. Equivariance ensures that replacing a prefix by the representative already chosen by the normalizer and type action does not alter this conclusion. The original acceptance tests and the original equivalence test still apply; this additional necessary condition accepts no new family.

One may start from an already complete saved level of the old search. Such a level may contain prefixes subsequently ruled out by this bound. They can be removed or skipped by the same argument. Earlier complete full-group families remain valid. Later level sizes describe the search with this exclusion, rather than the enumeration of all proper generated families. The distinction is retained in the execution records.

The computation forms the symmetric compatibility matrix using the unchanged rank-two test, then forms each actual compatible panel join. For a rejected prefix it records that prefix, its complete compatible index set and explicit generators for the proper join. Its correctness depends on the acquired complete panel set, complete earlier level when resuming, and exact finite-group operations. This is the previous generation bound applied after a necessary local restriction; no new classification or whole-source claim is made by the bound alone.

Transport of compatible panel sets

Human Compression project

Let $B \leq G$, let $\mathcal{P}_B = (P_1, \ldots, P_n)$ be the acquired complete ordered panel inventory, and let A be the image of $N_G(B)$ on its indices. The acquired rank-two test defines a symmetric relation $\sim$ on distinct indices. Conjugation preserves the subgroup and coset conditions in that test, so $\sim$ is A-invariant. Put $R(i) = \{j : i \sim j\}$.

Lemma 1. *Choose one representative r from each A-orbit. Testing one representative of each A_r-orbit determines $R(r)$ exactly. If $r^a = i$, then $R(i) = R(r)^a$. This formula is independent of the chosen transporter a.*

Proof. For $s \in A_r$, invariance gives $r \sim j$ if and only if $r \sim j^s$. Thus every point stabilizer orbit is either wholly accepted or wholly rejected. The singleton orbit $\{r\}$ is rejected by the distinctness convention. This proves the first assertion. Applying the same invariance with an element a sending r to i proves the second. Any two such transporters differ by an element of A_r, which preserves $R(r)$. □

For a selected family of indices T, the complete set used in the previous generation bound is consequently

$$C(T) = T \cup \bigcap_{i \in T} R(i).$$

The intersection over an empty family means all panel indices. A panel outside T belongs to this set precisely when it is compatible with every selected panel. As before, every admissible extension of T generates a subgroup of $J(T) = \langle B, P_j : j \in C(T) \rangle$. Thus $J(T) < G$ rules out every full-group extension. This is exactly the previous bound, with the same acceptance and equivariance proof.

The implementation stores the stabilizer orbits and the test result on each representative for each chosen r. It constructs other rows only when requested, using an actual permutation $a \in A$ with $r^a = i$. It records the transporter and the complete transported index set. No quadratic compatibility matrix is required. Membership in the actual action image, the transporter equation, and the equality of the transported set are explicit certificate checks. A rejected prefix still records its exact $C(T)$ and generators of its proper join.

The continuation differs from the previously reviewed compatible-panel continuation only in these row lookups and their serialization. The original rank-two, maximal augmentation, family acceptance and equivalence tests remain unchanged. A saved complete old level is still an acquired input. Subsequent level sizes count the filtered search, while all full-group families remain covered by the previous preservation argument. Joining the panels may stop once the join equals G, since every remaining panel is already a subgroup of G.

These are elementary orbit and transport arguments. Their use here does not assert a new classification. Complete panel inventories, the exact normalizer action, finite group operations and the acquired augmentation theorem remain explicit premises. The existing full-matrix computation provides a small independent comparison for the new representation.

Fusion of the completed Higman–Sims families

A certificate using the acquired full automorphism action

Let G be the retained permutation group on 100 points. The accepted normalizer certificate supplies $G \trianglelefteq N$, with $|G| = 44352000$, $|N| = 88704000$ and $C_N(G) = 1$. The acquired complete order $|\operatorname{Aut}(G)| = 88704000$ therefore identifies the conjugation action of N with the full automorphism group. This is an explicit acquired premise; the present calculation does not repeat an automorphism-group or subgroup census.

Proposition 1. *The simultaneous transporter in* `fusion_conjugation.g` *decides equivalence of ordered subgroup families under conjugation by N and arbitrary permutations of their types. Every positive answer contains a literal permutation of the 100 points and a type permutation.*

Proof. Fix a type permutation, so that the desired target family is $(Q_1, \ldots, Q_r)$. Begin with the original family $(P_1, \ldots, P_r)$, accumulated conjugator $a = 1$, and acting group $C = N$. At step i, find $x \in C$ carrying the current ith subgroup to Q_i; if none exists, this type permutation is impossible. Conjugate every current subgroup by x, replace a by ax, and replace C by $N_C(Q_i)$. The earlier target subgroups remain fixed. Inductively, every possible completion differs from this chosen partial transporter by an element of the new C: if two elements carry the same subgroup to Q_i, their quotient on the target side normalizes Q_i. Thus a failed step excludes every completion for this type permutation, and success carries the entire family. Trying every type permutation is exhaustive. Orders of individual groups, pair intersections and the full intersection are used only as necessary rejection tests. The code checks $a \in N$ and $P_i^a = Q_i$ literally before returning. □

The completed anchor inventory covers every one of the 589 acquired subgroup classes exactly once: 529 by completed enumeration and 60 by the separately proved proper-generation bound. The latter have no full-G families and are recorded separately, not represented by invented empty enumerations. For each retained full family of panels $(T_1, \ldots, T_r)$ with Borel B, form $P_i = \langle T_j : j \neq i \rangle$. The acquired intersection property gives $\bigcap_i P_i = B$ and $T_i = \bigcap_{j \neq i} P_j$; the calculation also checks these equalities on each actual family.

Successive representatives are compared by the proposition. A failed comparison separates two orbits; a positive witness assigns the new family to an existing orbit. This partitions precisely the completed input. The output concerns geometries together with their specified G action, modulo $\operatorname{Aut}(G)$ and types. Identification with the source's bare incidence classes requires its separate isomorphism comparison and is not inferred from this partition alone.

The transporter is the earlier simultaneous-family method written in the already acquired permutation action, with its actual conjugator retained. Small controls compare it with exhaustive enumeration of all acting permutations and all type permutations, including the outer action of S_4 on A_4. The complete target and every input receipt remain separately bound. No manuscript, conjecture or Lean credit is assigned by this method note.

Incidence invariants for the Higman–Sims geometries

L091 classification component

Let $G = \mathrm{HS}$ in the acquired permutation representation of degree 100. Its order is 44 352 000. We use the complete subgroup and panel inventory and the separately reviewed enumeration of the firm, residually connected, flag-transitive, residually weakly primitive, locally two-transitive IP2 coset geometries for this action. In ranks at least two, that enumeration gives 130 families and 86 classes under $\mathrm{Aut}(G)$ and permutations of the types. The rank counts are respectively $(6, 44, 67, 12, 1)$ and $(6, 30, 41, 8, 1)$ for ranks two through six. There are no higher ranks. The acting automorphism group is represented faithfully by the acquired normalizer of order 88 704 000.

The enumeration and its completeness are inputs here. In particular, the present argument must still exclude an incidence isomorphism which does not preserve the specified action of G. We refer to the numbered representatives in the fixed, acquired file `full_fusion03/result.g`. These numbers are internal identifiers, not the source's diagram labels.

Lemma 1. *Suppose a group G acts transitively on each type of a finite typed incidence structure. Let X be a type and let $S_y \subseteq X$ be the set of objects incident with an object y of another type. If $G_{S_y} = G_y$, the map $gy \mapsto gS_y$ is injective. More generally, suppose the objects of each remaining type can successively be identified by their sets of neighbours in types already identified. Then restriction to X is faithful on the group of automorphisms preserving these types.*

Proof. The fibres of the equivariant map $G/G_y \longrightarrow G/G_{S_y}$ have size $[G_{S_y} : G_y]$. Equality of the stabilizers proves the first claim. An automorphism fixing X pointwise fixes every distinct subset of X, hence fixes the next type pointwise. Induction proves the second assertion. This argument does not require that G be the full automorphism group. □

Lemma 2. *Let Γ and Γ' have a distinguished type X and X', recognizable from their type sizes. Suppose each structure defines a graph on that type purely by incidence, restriction of its automorphism group to that type is faithful, and the image of its specified group G is the derived subgroup of the full graph automorphism group. Every incidence isomorphism $\Gamma \longrightarrow \Gamma'$ conjugates the specified G actions.*

Proof. An incidence isomorphism f induces a graph isomorphism f_X. It conjugates the full graph automorphism groups and consequently their derived subgroups. For $g \in G$, choose $h \in G$ whose action on X' equals that of fgf^{-1}. The automorphism $h^{-1}fgf^{-1}$ is trivial on X', so faithfulness makes it trivial on all of Γ'. This gives the asserted conjugation of the entire actions. □

Theorem 1. *Under the acquired enumeration and the exact computational inputs specified below, the 86 representatives are pairwise nonisomorphic as incidence geometries, even allowing permutations of types. There is one additional isomorphism class in rank one, consisting of a set of 176 objects. It is a truncation of a higher rank geometry. Thus the numbers of isomorphism classes in ranks one through six are*

$$1, \quad 6, \quad 30, \quad 41, \quad 8, \quad 1.$$

Proof. Write G_i for the stabilizers of a base chamber. Since the action is flag-transitive, the number of flags whose set of types is a nonempty subset J is $|G|/|\bigcap_{j\in J} G_j|$. These counts are incidence invariants, with their coordinates permuted when the types are permuted. In particular the multiset of orders $|G_i|$, the multiset of orders $|G_i \cap G_j|$, and the Borel order are invariants for these fixed-order actions. Computing these from the actual subgroups separates all representatives except the pairs $(14, 15)$ and $(83, 84)$. For the rank five pair $(83, 84)$, the vector of all nonempty flag stabilizer orders, minimized over every permutation of the five types, has different values. These are checked by intersections of the acquired subgroups, not inferred from the printed diagrams.

It remains to separate 14 and 15. Their three stabilizer orders are $192, 1920, 40320$. Their type sizes are therefore $231000, 23100, 1100$, so every isomorphism preserves each type. Call these types lines, blocks and points, respectively, and let P, Q, R denote their stabilizers at a base chamber. Take the actual action on the 1100 right cosets of R. It is faithful. If $z = R$, the shadows of the base line and block on the point type are z^P and z^Q, of sizes two and sixteen.

Define two points to be adjacent when some line is incident with both. This graph is intrinsic to the geometry. To construct it, let w be the other member of z^P. Its neighbours at z are exactly w^R: the lines incident with z form the R orbit of the base line. Transporting this neighbour set by a representative of every coset gives the whole graph. In both cases it has 1100 vertices and valency 105.

The stabilizer in G of the base block shadow z^Q has order 1920, equal to $|Q|$. Thus blocks are determined by their sixteen point shadows. Lines are *not* determined by their two point shadows: the corresponding fibre has size four. Instead consider the set $\mathcal{B} = (z^Q)^P$ of block shadows neighbouring the base line. There are four. The group P is transitive on them, and the intersection index $[P : P \cap Q] = 4$ confirms that they are all its incident blocks. Exactly 48 of the 1920 elements of Q stabilize $\mathcal{B}$ setwise. Put $T = G_{\mathcal{B}}$. We have $P \leq T$, and the orbit of z^Q under T is all four members of $\mathcal{B}$, since P is already transitive. Also $T_{z^Q} = T \cap Q$, by the block stabilizer equality. Hence

$$|T| = 4 \cdot 48 = 192 = |P|.$$

The four neighbouring blocks therefore determine each line. Lemma 1 proves faithfulness of restriction of the full geometry automorphism group to the point type.

The complete graph automorphism computation returns order 88 704 000 for each of these two graphs. Every returned generator is checked against every adjacency. In GAP the group D generated by these permutations has this exact order, contains the actual coset image A of G, and satisfies $D' = A$. Completeness of the graph computation is the explicitly retained nauty input; the subgroup equality $D' = A$ is tested on the actual permutations. Lemma 2 now shows that any incidence isomorphism between these geometries would conjugate their full specified G actions. Such an isomorphism induces an automorphism of the abstract group G. After composing with an element of G to move the image chamber back to the base chamber, it sends the unordered family of parabolic stabilizers onto the other family. The acquired complete $\mathrm{Aut}(G)$ comparison excludes this for representatives 14 and 15. This proves that all 86 representatives are distinct.

Finally, the complete panel inventory has exactly two two-transitive actions in rank one, at subgroup anchors 586 and 587. Both completed enumeration records have levels $(1, 0)$, and both actions have degree 176, are faithful and are two-transitive. A rank one geometry has no incidence between distinct types, so its isomorphism type depends only on its number of objects. It has full automorphism group Sym(176). The retained conjugators in the acquired normalizer identify each point stabilizer with a parabolic of a higher rank representative. They give explicit truncation isomorphisms. There is consequently one rank one class, and none which is not a truncation. Together with the acquired exhaustion in higher ranks, this proves the theorem. □

Attribution and limits. The counts recover the classification asserted by Dimitri Leemans, *The residually weakly primitive geometries of HS*, Australasian Journal of Combinatorics 33

(2005), 231–236, in the fixed local six-page edition. The complete subgroup catalogue, the panel enumeration and the full automorphism action retain the attributions and computational premises of their acquired reviews. The shadow lemmas are elementary consequences of equivariance, orbit–stabilizer and characteristic subgroups. Their application above supplies the missing distinction between equivalence of group actions and isomorphism of the geometries; it does not assert a new classification or a solution to a new open question.

The exact retained computational inputs are GAP's permutation group, coset action and subgroup algorithms and nauty's complete graph automorphism algorithm. The supplied generators, adjacency tests, stabilizer and derived subgroup equalities are additional finite checks, not a substitute for nauty completeness. No Lean formalization is claimed. The higher rank truncation counts, the supplementary diagrams and full automorphism data, and the source's two Hoffman–Singleton constructions remain separate obligations. This component therefore gives no whole-manuscript completion or compression measurement.

Truncations of the Higman–Sims geometries

L091 classification component

We use the complete enumeration and the separate proof of bare incidence isomorphism classes for the specified Higman–Sims action. There are $1, 6, 30, 41, 8, 1$ classes in ranks one through six and no higher ranks. A truncation below means deletion of types from another geometry in this classified family. The internal representative numbers refer to the fixed file `full_fusion03/result.g`.

Proposition 1. *Exactly two rank two classes and ten rank three classes are truncations of higher rank geometries in the family. No class of ranks four, five or six is such a truncation. The sole rank one class is a truncation. Consequently the numbers of classes which are not truncations are*

$$0, \quad 4, \quad 20, \quad 41, \quad 8, \quad 1.$$

Proof. Let G have order $44\,352\,000$, and let $(G_i)_{i\in I}$ be a base chamber's parabolic stabilizers. For a subset $J \subseteq I$, the truncation is the coset geometry with parabolics $(G_i)_{i\in J}$. It inherits flag transitivity. Thus its type sizes, pair-flag counts and chamber count determine respectively the multisets

$$\{|G_i| : i \in J\}, \qquad \{|G_i \cap G_j| : \{i, j\} \subseteq J\}, \qquad \left|\bigcap_{i\in J} G_i\right|.$$

These are necessary invariants for incidence isomorphism with any member of the classified list, since the group order is fixed. Their use does not assume that an incidence isomorphism normalizes G.

For every one of the 86 representatives of rank at least two, take every proper subset J with $|J| \geq 2$. There are exactly

$$30\binom{3}{2} + 41\left(\binom{4}{2} + \binom{4}{3}\right) + 8\left(\binom{5}{2} + \binom{5}{3} + \binom{5}{4}\right) + \binom{6}{2} + \binom{6}{3} + \binom{6}{4} + \binom{6}{5} = 756$$

such subsets. For each, compare the displayed invariant with every representative. Exactly fifteen comparisons survive. Each has a saved permutation in the acquired full automorphism action of G and a permutation of types, sending the actual truncation parabolics onto those of its target. These give explicit coset isomorphisms. There are no surviving comparisons without a positive witness. The distinct targets are $3, 4$ in rank two and $8, 13, 18, 20, 21, 22, 28, 29, 30, 31$ in rank three.

An arbitrary classified parent is incidence-isomorphic to a listed representative; any such isomorphism transports its type subsets. Hence the finite list of comparisons covers every possible parent and truncation. Every excluded comparison fails an intrinsic incidence invariant, while every surviving one is positively identified. This proves completeness of the claimed twelve targets, including the negative assertions in ranks four through six. The separately checked rank one component supplies a truncation witness for its unique 176-object geometry. Subtraction from the complete rank counts proves the displayed result. □

Attribution and computational scope. These are the truncation totals stated in D. Leemans, *The residually weakly primitive geometries of HS*, Australasian Journal of Combinatorics 33 (2005), 231–236. Completeness of the parent classification is an explicit acquired premise.

The new finite comparison uses actual subgroup intersections and the acquired simultaneous conjugation method in GAP; every positive witness is checked directly. No inference uses failure of a conjugator search. This component supplies the truncation clause, not the supplementary diagrams, automorphism groups or the two geometric constructions, and claims no new open result.

Local data of the Higman–Sims coset geometries

We acquire the reviewed classification of 86 coset geometries for the literal faithful group $G \cong$ HS, of order 44 352 000. The numbers in ranks two through six are $6, 30, 41, 8, 1$. The acquired action is flag-transitive. Its parabolics are denoted by G_i, $i \in I$, and $G_J = \bigcap_{j \in J} G_j$, with $G_\varnothing = G$. The Borel subgroup is $B = G_I$. All identifiers below refer to the fixed representatives in `full_fusion03/result.g`.

Theorem 1. *The retained finite calculations determine, for each of these 86 representatives, its type sizes, every flag stabilizer order, its Borel and panel orders, its node orders, and the orders and literal generators of its local action kernels. They also determine the two ordered diameters and girth of each of its 437 rank-two residues at a base flag. Finally, they determine the normalizer of the specified G action in the type-preserving automorphism group. For representatives 14 and 15 this is the full automorphism group, of respective orders* 44 352 000 *and* 88 704 000.

The assertion is a computation of the actual models. It does not assume agreement with any printed group name or diagram label. The historical supplement of 6 October 2003, attributed to Dimitri Leemans, is compared with these values in a separate source audit. No new classification or open-conjecture resolution is asserted here.

Stabilizers and local kernels

The objects of type i are the right cosets of G_i, so their number is $|G : G_i|$. The stabilizer of the standard flag of type J is exactly G_J. Flag transitivity gives $|G : G_J|$ flags of that type. Fixing the standard flag of cotype $\{i\}$ leaves $|G_{I\setminus\{i\}} : B|$ objects; consequently the node order is this index minus one.

Fix an object of type i. For $j \neq i$, its incident objects of type j form the transitive G_i-set with stabilizer $G_i \cap G_j$. The kernel on that set is the core of this intersection in G_i. An element acts trivially on the whole residue precisely when it acts trivially on every type. Thus the local kernel is

$$K_i = \bigcap_{j \neq i} \mathrm{Core}_{G_i}(G_i \cap G_j).$$

The calculation uses these actual intersections and cores. It saves generators of B, the panels, each K_i, and the original parabolics, as well as their orders. An order by itself is not a structural group identification.

An indexed construction of the incidence graphs

For a residue of cotype $\{i, j\}$ put $H = G_{I\setminus\{i,j\}}$, $P = H \cap G_i$, $Q = H \cap G_j$, and $D = P \cap Q = B$. Its two types are the cosets of P and Q in H. Take right transversals T_P, T_Q in H and a right transversal T_D of D in P. For $x \in T_P$, its full neighbour set is

$$\{Qux : u \in T_D\}. \tag{1}$$

Indeed a coset meeting Px contains some px, $p \in P$, and is therefore Qpx. Two such cosets agree precisely when $p_1 p_2^{-1} \in P \cap Q = D$. Thus (1) contains each neighbour exactly once. A

position lookup in T_Q gives the integer vertex label without constructing a large permutation action.

For the six rank-two geometries this construction produces 158,447 vertices and 537,835 edges in total. The completed invocation took 34.36 seconds. An earlier generic action construction reached its 120-second bound without returning a graph; that record is preserved. The new construction changes the indexing method, not the incidence relation. Independent small controls compare every edge with literal coset intersection.

For higher ranks, 425 graphs were obtained by transporting the base P-orbit in the action of H on Q-cosets. This is the same neighbour formula. Six graphs for representatives 14 and 15 are acquired from their earlier reviewed comparison. Together the 437 graphs have 187,005 vertices and 601,049 edges. The exact roster contains one graph for every unordered type pair of every representative. The reader checks every adjacency, both valencies, the two flag indices, and $|H : B|$ edges. These checks supplement the exact coset construction; they do not substitute matching counts for incidence.

Distances and the diagram parameters

Lemma 1. *In a finite connected bipartite graph containing a cycle, whose automorphism group is transitive on each part, breadth-first search at one vertex in each part determines both type diameters and the girth.*

Proof. The eccentricity is constant on each part by transitivity. Hence the largest distance found from its chosen vertex is that type's diameter. For a breadth-first tree rooted at v, a non-tree edge xy gives the number $d(v, x) + d(v, y) + 1$. Its tree paths and the edge contain a cycle of length at most this number, so every such number is at least the girth. Choose a shortest cycle and move a vertex of it to v using transitivity on the appropriate part. Its length is even, say $2m$. There is no shorter path between vertices of this cycle than the shorter arc, since such a path would give a shorter cycle. The vertex opposite v therefore has distance m, and its two cycle neighbours have distance $m - 1$. At least one of these two edges is not a tree edge and gives $2m$. The minimum recorded number is consequently the girth. This argument applies to either part. □

The acquired coset action is transitive on each part. All valencies are at least two, so these finite graphs contain cycles and the lemma applies. The two searches must agree on the girth. The diagram triple is the left type diameter, half the girth, and the right type diameter. A digon is checked as a complete bipartite graph. A complete-graph residue is recognized only when one type consists of all distinct two-subsets of the other, with exactly that incidence.

Normalizers and the two remaining ambiguous profiles

Let $N \cong \mathrm{Aut}(G)$ be the acquired faithful automorphism action on G, and put $S = \bigcap_i N_N(G_i)$. Let M be the normalizer of the specified G action in the type-preserving geometry automorphism group. The chamber stabilizer M_Φ maps by conjugation into S. This map is injective: an element fixing Φ and commuting with G fixes every chamber, because G is chamber-transitive, and hence fixes every object. Conversely each automorphism in S induces the chamber-fixing coset map $G_i x \mapsto G_i \alpha(x)$. Thus $M_\Phi \cong S$. Since $G \le M$ is chamber-transitive,

$$|M| = |G : B|\,|S|.$$

This formula computes a normalizer; it does not in general prove that all geometry automorphisms normalize the given copy of G.

For representatives 14 and 15, that last assertion is already paid by the reviewed shadow argument. Each defines a graph on its 1100-point type; restriction to that type is faithful, and the derived subgroup of the full shadow-graph group is precisely the specified G. Every geometry automorphism therefore normalizes G. Their Borel orders are both 24, while the calculated orders of S are respectively 24 and 48. The displayed formula gives the two full orders stated above. All three type sizes are different, so their correlation groups preserve types as well. For representative 15 the resulting group is the full shadow-graph group, which is isomorphic to $\mathrm{Aut}(G)$ by its faithful conjugation action. Representative 14 has full group equal to the specified G itself. To justify the faithful conjugation assertion, the calculation also checks $S \cap G = B$ in both cases. An automorphism commuting with G takes the base chamber to Φ^g for some $g \in G$; equality of the object stabilizers forces $g \in S \cap G = B$. It therefore fixes the base chamber and, by transitivity, the entire geometry. The conjugation kernel is trivial.

Attribution and computational boundary. The group model, complete family classification, full automorphism action and faithful shadow argument are acquired reviewed results. The general arguments here are elementary coset and group-action facts; their explicit derivations, numerical applications and validation programs are supplied here. Exact GAP permutation-group algorithms remain computational premises. No historical source label, full automorphism claim for the other 84 models, abstract identification of local kernels, Lean verification, measurement or whole-manuscript completion follows from this component alone.

Coordinates and residues in the Higman–Sims catalogue

Private L091 component

Let $G = \mathrm{HS}$. We use the supplement dated 6 October 2003, with 23 pages, as the fixed source for the diagram numbers below. Its statements and notation remain attributed to the original authors. The complete classification, the local numerical data, the bare isomorphism distinctions, and the truncation certificates are acquired from their separately reviewed components. This application identifies the source coordinates and verifies the small named residues. It does not replace those earlier proofs.

Theorem 1. *The numerical profiles of the 86 diagrams of ranks two through six, with the six explicit amendments below, give a bijection to the 86 classes in the acquired classification. The thirteen printed truncation assertions have actual incidence isomorphisms. The thirteen occurrences of a Petersen residue and the two occurrences of an affine-plane residue have explicit isomorphisms to the Petersen incidence structure and* $\mathrm{AG}(2,4)$*, respectively.*

Coordinates and the two ambiguities. For each source diagram, first compare the multiset of type sizes with every acquired class. Apart from two pairs, this identifies a unique class. The entry 44320 in diagram 2.6 is retained in the transcription but replaced by 44352 for this comparison: the printed integer does not divide $|G| = 44352000$, so it cannot be an index for the asserted transitive action.

The two rank-three diagrams 3.22 and 3.23 have identical type sizes and local numerical data. Their candidate classes are 14 and 15. The acquired full automorphism proof gives $\mathrm{Aut}(\Gamma_{14}) = G$ and $\mathrm{Aut}(\Gamma_{15}) = \mathrm{Aut}(G)$, of respective orders 44352000 and 88704000. These are full typed groups, rather than merely normalizers of a specified action. Thus the source's two different automorphism assertions select 14 for 3.22 and 15 for 3.23. For the remaining tied pair, 5.2 and 5.3, the complete diagram has respectively one and two quadrangle bonds. Every permutation among equal-sized types is checked. This selects classes 83 and 84.

The finite coordinate list records a permutation from each source's ordered nodes to the actual model's types. The resulting 86 class indices are precisely $1, \ldots, 86$, each once. This identifies classes by proved invariants within a complete classification; it makes no claim that an unavailable historical generator list is literally the same list of permutations as ours.

Literal discrepancies. An ordered triple (a, g, b) records the eccentricities of the two specified vertex types and half the girth of their incidence graph. The finite record retains both the printed and the actual values. The six amendments are as follows.

Diagram	Amendment
2.5	The node order at type 36960 is 15, in place of 11.
2.6	The second type has size 44352, in place of 44320.
3.15	The right arm, ordered from the type of size 352 to that of size 1100, has triple $(5,4,6)$, in place of $(6,4,5)$.
3.30	The top node order is 3, in place of 1.
4.22	The bottom bond is a digon; the omitted diagonal joining the two types of size 352 has a single bond.
4.28	The bond from type 352 to type 1386000 has triple $(7,5,8)$; the bond from type 1386000 to type 1100 has triple $(4,3,3)$ and is a complete-graph incidence structure.

Why these are amendments. For each of these six rows the type-size multiset, after the explicitly declared index amendment in 2.6, selects a unique class. All compatible type permutations are tested against the acquired local data. Therefore the stated mismatch cannot be removed by choosing another class or by silently exchanging equal-sized types. In particular, both possible exchanges of the 352 types

in 4.22 remain inconsistent with its printed bonds. The amended values are consequences of actual residue graphs, not an inference from visual similarity to another diagram.

The independent source audit also records order inconsistencies in the displayed parabolic, Borel or kernel names of 3.1, 3.4, 3.5, 3.12, 3.13, 3.15, 3.16, 3.20, 4.24 and 4.28. We retain those discrepancies. This theorem does not supply the abstract names that should replace them, nor prove the remaining names merely because their orders agree. The source's boxed kernel and generic extension dot keep their own meaning; the dot alone is not treated as an assertion of nonsplitting. The Aut and Cor values left blank in 3.10 are still recorded as blank.

Truncations. Restricting a coset geometry to a subset J of its types retains the same cosets and the same incidence relation. Each previously reviewed positive truncation certificate gives an automorphism α of G and a type permutation carrying the retained parabolics to those of the target. The maps

$$G_i x \longmapsto \alpha(G_i)\alpha(x)$$

are well defined bijections. Two cosets meet if and only if their images meet, so these are incidence isomorphisms. Composing the already verified certificates with the coordinate list above proves all thirteen printed links. The record gives the exact retained source nodes for each link. No new transporter search is needed. The eight printed doubling assertions are separate obligations.

Petersen and affine residues. Take as Petersen vertices the ten two-element subsets of a five-element set, with adjacency defined by disjointness. The incidence structure has these ten vertices as points and its fifteen graph edges as lines. This is the standard Petersen model used in the thirteen certificates.

For the affine model, set $F = \mathbb{F}_2[t]/(t^2 + t + 1)$. The polynomial has no root in $\mathbb{F}_2$, so F is a field of order four. The points are F^2; its twenty lines are $y = ax + b$ and $x = b$. Two distinct points lie on a unique such line, by division of their coordinate difference when the first coordinates differ, and by the vertical line otherwise. Thus this is $\mathrm{AG}(2, 4)$.

For each of the fifteen source occurrences the certificate supplies a bijection from the full displayed model to the actual residue graph. For every vertex, the image of its entire neighbor set equals the entire neighbor set of its image. This proves incidence preservation in both directions. The model points are also checked to occupy one entire actual type. Consequently the record gives the exact point polarity, while retaining every printed star; no unstated convention about reading stars on sloping diagram bonds is imposed.

Evidence and scope. The source transcription was checked on all 23 native pages by two readers. The new coordinate calculation uses the accepted complete finite records. Graph software only proposed the fifteen small positive maps; their verification uses literal adjacency lists and the explicit models above. Damaged map, roster and coordinate controls accompany ordinary and optimized executions. The finite inventory and each acquired review are fixed by exact file hashes. This component contributes no new census, whole-original completion, conjecture resolution, formalization or compression measurement. The source transcription, earlier classification and earlier local-data proofs retain their separate authorship and review records.

Borel and residue-kernel atoms in the Higman–Sims catalogue

Private L091/L096 finite component

The fixed source is Dimitri Leemans's 23-page Higman–Sims supplement, dated 6 October 2003, retained with the L096 materials. Diagram numbers refer to that edition. We acquire the reviewed 86-family classification, its literal source-coordinate bijection, and the computed Borel and residue-kernel generators. This note supplies a first structural batch, without another subgroup or geometry census.

Theorem 1. *The 350 occurrences in the literal source-join ledger have the structures listed below. They comprise 55 Borel groups and 295 residue kernels. Among these occurrences, the only changes to the printed group labels are the Borels of diagrams 3.15 and 3.16: both are* $C_3^2 : Q_8$*, in place of the printed* $2^4 : A_4$*.*

Structure	Borels	Kernels
A_4	1	0
A_5	2	1
1	0	204
C_2	1	26
C_3	3	5
C_4	4	5
C_4^3	0	1
$C_5 : C_4$	3	5
C_6	1	1
D_{12}	3	0
D_8	9	0
C_2^2	10	12
C_2^3	0	11
C_2^4	0	12
$C_3^2 : Q_8$	5	1
$L_3(4)$	1	1
Q_8	7	8
S_4	2	0
S_6	3	2
Total	55	295

Here D_n denotes the dihedral group of order n, and powers of C_m denote direct powers, not an unspecified group of the same order. The remaining 31 Borels and 17 kernels are listed individually in `remaining.json`. No parabolic structure or quotient structure is asserted merely from one of these kernel identifications.

The actual subgroup interface. For the acquired right-coset geometry let G_i be its standard parabolics. The Borel is $B = \bigcap_i G_i$. The residue kernel at type i is

$$K_i = \bigcap_{j \neq i} \mathrm{Core}_{G_i}(G_i \cap G_j).$$

The reviewed local-data component proves this formula and records the actual generators of these groups on the same 100 points. We use those generators literally. The source-coordinate permutation specifies which model type belongs to each printed node; equal orders do not choose that association. The source's box denotes the kernel even when it occurs inside parentheses, as in $(\boxed{2} \times A_6) \cdot 2^2$. The dot separating a kernel and quotient is generic extension notation; it is not an additional nonsplitting assertion.

Lemma 1 (Literal generator-edge transport)**.** *Let $M = \langle s_1, \ldots, s_t \rangle$ and $U = \langle u_1, \ldots, u_v \rangle$ be finite permutation groups. Suppose a map $f : M \longrightarrow U$ is a bijection, $f(1) = 1$, and for fixed $h_i \in U$ it satisfies $f(xs_i) = f(x)h_i$ for every $x \in M$ and every i. Then f is an isomorphism carrying s_i to h_i.*

Proof. Every element has a word in the generators and their inverses. Since the groups are finite, inverse powers can also be replaced by positive powers. Induction on a word for y gives $f(xy) = f(x)f(y)$. Thus f is a bijective homomorphism. This is the acquired marked-coordinate transport argument, here applied to the small literal subgroup closures. □

The certificate records the images of the ordered model generators. The reader computes the entire model closure and every right-generator edge, propagating images from the identity. Every repeated encounter must give the same image. It independently computes the actual subgroup closure, checks image membership, and compares the complete element sets bijectively. Thus the generator images are compact data for a full finite certificate, not an unverified isomorphism proposal. The 350 checks use 92,530 edges.

Elementary models. Cyclic and homocyclic groups use disjoint cycles. Their full distinct normal forms are checked; in particular C_4^3 is not inferred from order 64. Dihedral models use rotation and reversal on a polygon. The quaternion model satisfies $a^4 = 1$, $b^2 = a^2$, and $b^{-1}ab = a^{-1}$, with the eight distinct forms $a^i b^j$, $0 \leq i < 4$, $0 \leq j < 2$. For S_n and A_n, $n \leq 6$, the generated element set is compared with all permutations, respectively all even permutations, of the specified points. The order-20 model is the affine group of the five-element field: translations and multiplication by 2 give an actual $C_5 : C_4$ decomposition. The sole non-elementary named atom, $L_3(4)$, is acquired from the accepted L100 projective construction through the reviewed M22 and linear-table registry. Its exact 21-point generators are checked, not replaced by an order-based recognition. The named projective-group interface remains an explicit inherited premise.

The five Borels of order 72. For completeness we specify the relevant affine model directly. On $V = \mathbf{F}_3^2$, use the two coordinate translations together with the right linear actions of

$$A = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

They satisfy $A^2 = C^2 = -I$ and $AC = -CA$. Their eight distinct products $A^i C^j$ form Q_8. The linear group fixes the zero vector, whereas the translation group acts regularly. Hence their intersection is trivial, translations are normal, and their product is $V \rtimes Q_8$ of order 72. The reader checks all these assertions on the nine actual points, including the distinct normal forms and nontrivial complement action. It also checks that every nonidentity element of the complement fixes no nonzero vector.

The literal transport certificates identify the Borels of diagrams 3.12, 3.13, 3.14, 3.15 and 3.16 with this model. The corresponding fixed representatives are 24, 21, 23, 25 and 22. The kernel of source type 2 in 3.14 (model type 1 in representative 23) has the same identification. The first three printed Borel labels already say $3^2 : Q_8$. The last two instead say $2^4 : A_4$, which has order 192. Their native pages 6 and 7 have been inspected directly. Thus the correction is supported both by the acquired exact order and by a positive structural certificate. It does not alter the principal geometry classification.

Source checks, validation and limits. The full ledger records the source page, diagram, printed label, source node, model node, actual order and proved model for each occurrence. The source transcription and coordinate theorem are acquired from their separate reviews. A native recheck of pages 7–8 caught an initial extraction mistake: four boxes inside parentheses had been incorrectly treated as absent. The predecessor diagnostic is preserved. Correct extraction agrees with all four printed C_2 kernels; no source correction is asserted there.

One capped GAP invocation proposed the finite maps in 2.26 seconds. Its receipt and all proposals are retained. The final proof reader calls neither GAP nor an isomorphism oracle. Its ordinary and optimized runs must agree. Damaged map images, wrong subgroup orders and generators, an incorrect same-order model, an elementary substitute for C_4^3, malformed metadata and interior-box examples exercise the certificate boundaries. An initial negative control expected a later bijection failure but correctly failed earlier at a generator edge; that harness failure is preserved.

The work acquires prior mathematical classifications and literal models; it does not reprove those classifications. The new model checks, transport application, source extraction and program are by the present component's author. Root assigned the scope and the two known order discrepancies; the atlas author supplied registry pointers. No whole-source, new question, formalization or measurement credit is claimed. Compound groups outside the 350-occurrence ledger, all parabolic structures, other sporadic source clauses, and external rank-six construction attributions remain separate.

The remaining compound local groups in the Higman–Sims catalogue

Private L091/L096 finite component

We use the fixed 23-page supplement of Dimitri Leemans, dated 6 October 2003. Its diagram coordinates are those in the acquired source-coordinate ledger, not positions inferred from group orders. The reviewed local-data component supplies literal generators on 100 points for

$$B = \bigcap_i G_i, \qquad K_i = \bigcap_{j \neq i} \mathrm{Core}_{G_i}(G_i \cap G_j).$$

The earlier local-atom component identifies 350 of the 398 Borel and kernel occurrences. This note identifies the remaining 31 Borels and 17 kernels. It does not identify the full parabolics or their extensions over these kernels.

Theorem 1. *All 48 occurrences in* `source_joins.json` *have the displayed structures and the literal permutation models recorded there. Nine printed occurrences require correction, as specified on page 3. Together with the earlier 350 certificates this identifies all 86 Borels and all 312 residue kernels of the acquired catalogue.*

Finite transport and acquired names. The finite certificate gives images of the ordered generators of a named model M in an actual subgroup U. The verifier closes both permutation groups and constructs a bijection $f : M \to U$ fixing identity. For every element x and every model generator s it checks $f(xs) = f(x)f(s)$. Induction on positive words proves that f is a group isomorphism; inverses are positive powers because the groups are finite. This is the previously acquired generator-edge transport argument. The names of the reused subgroup-atlas models are acquired with their exact trees, degrees and generators. An order match alone is never used as a named isomorphism. New elementary factors are the literal cyclic, dihedral, quaternion, symmetric, alternating and affine models already proved in the first local-atom component. Direct products act on disjoint supports. The generalized dihedral model of order 50 acts on $\mathbb{F}_5^2$ by its two translations and $v \mapsto -v$.

Eight additional models have a recursive certificate $U = N : C_p$, with $p = 2$ or 3. A full literal map identifies N with an earlier model. The supplied element of order p lies outside N, normalizes it, and the verifier checks that the products nt^i are exactly U. Consequently the claimed split extension follows with its actual conjugation action. The recursive dependency order is checked; no model is justified by a circular reference. All maps and recursive factors are verified by `compound.py`, which makes no group-software or isomorphism call.

Reconciling different decompositions. A different extension tree does not by itself contradict a source label. We therefore supply ten additional normal-subgroup/complement certificates. Full generator-edge maps identify both factors, the normality relations are checked, their intersection is trivial, and their set product is the whole group. These certificates yield the following source-facing descriptions; $E_{p^d} = C_p^d$.

Base occurrence	Normal factor	Complement
2.2 Borel	C_4^3	S_4
3.6 Borel	E_8	S_4
3.10 Borel	E_{16}	A_4
3.20 Borel	E_{16}	C_2
3.24 Borel	Q_8	C_2
3.25 Borel	Q_8	C_2
4.2 Borel	E_{16}	E_4
4.7 Borel	E_{16}	C_2
4.12 Borel	E_4	E_4
4.11 Borel	E_9	C_2

For any second occurrence identified with the same exact model, compose its isomorphism with the inverse of the first. Transporting the factor generators gives the same decomposition in the second actual subgroup. The verifier checks these transported factors again, giving 32 factor checks at 28 distinct occurrences. In particular it pays the repeated $2^3 : S_4$, $2^4 : A_4$ and $2^4 : 2^2$ names. The order-40 Borel in 3.19 has an additional direct certificate $C_2 \times \mathrm{AGL}(1,5)$: the cyclic factors of orders 5 and 4 have the literal affine action, and the central involution is an independent factor.

The Borels in 3.25 and 3.26, and the boxed kernel in 3.26, are more precisely $Q_8 \times C_2$. They are split extensions written $Q_8 : 2$ in the source; the complement action is trivial. The Borel in 4.7 also splits. The source's dot is retained as extension notation here; we do not turn it into an unprinted nonsplitting assertion.

Lemma 1 (The common central kernel of order 64)**.** *The kernels in 2.2, type 2; 3.4, type 3; and 3.5, type 1 are central nonsplit extensions $C_4 \cdot E_{16}$.*

Proof. For the first actual kernel the certificate gives a central element z of order 4 and four quotient lifts. Their squares and commutators lie in $\langle z \rangle$. The sixteen binary words in these lifts give sixteen distinct cosets covering the group. The quotient is therefore E_{16}. Two supplied lifts do not commute. Multiplying either lift by a central element leaves their commutator unchanged; the verifier also checks all sixteen such pairs. An elementary abelian complement would provide commuting lifts of every two quotient elements, a contradiction. The other two kernels are explicitly isomorphic to this one by their common exact model. This proves the three printed $4 \cdot 2^4$ labels with the stronger central and nonsplit qualification. □

The nine incompatible printed occurrences. Let X denote the actual extraspecial group of order 32 in acquired model 9011, with its literal generators. Let R denote acquired model 21012, an extension $X \cdot E_4$ of order 128. The indicated actions, not just the extension shapes, are fixed by the serialized models.

Source occurrence	Printed label	Proved description
2.3 Borel	$4^3 : (7 : 3)$	$E_8 : L_2(7)$
3.4, 3.5 Borels	$(4^3 : 2^2) : 2$	$R : C_2$, order 256
3.4, type 2 kernel	$4^3 : 2^2$	$(((D_8 \times C_2) : C_2) : C_2) : C_2$, order 128
4.2, 4.3, 4.4, type 2 kernels	$2^4 : 2$	X
4.25 Borel and type 2 kernel	D_8	$C_4 \times C_2$

Here D_8 has order 8. These are nine separate entries in seven diagrams. The source coordinates and their literal model-type permutations are retained in the ledger.

For 2.3, four explicit commutators generate the actual group of order 1344, so it is nontrivial and perfect. The printed $4^3 : (7 : 3)$ is soluble: both its abelian normal factor and its group of order 21 are soluble. Thus the literal isomorphism to $E_8 : L_2(7)$ is a genuine correction, although the two displayed orders agree. For 3.4 and 3.5 the printed orders are respectively 256 for the kernel and 512 for the Borels; the actual certified orders are 128 and 256. The recursive positive models give structures as well as these order obstructions.

For each of the three order-32 kernels, transport identifies it with X. Direct checking gives $Z(X) = X' = C_2$, and every square lies in the centre. The commutator pairing on $X/Z(X) \cong \mathbb{F}_2^4$ is alternating and nondegenerate: a radical element would lift to the centre. If $A \leq X$ is abelian, $AZ(X)/Z(X)$ is isotropic, so has dimension at most 2. Hence $|A| \leq 8$. In particular X has no elementary abelian subgroup of order 16 and cannot have the printed $2^4 : 2$ structure. Finally, the two 4.25 groups have the explicit abelian model $C_4 \times C_2$, whereas D_8 is nonabelian.

Validation and scope. The new replay verifies all 48 occurrences and 48,647 generator edges, the stated factor certificates and the three obstruction arguments. Fourteen damaged certificates are rejected in both ordinary and optimized Python. Finite proposal calls and two unsuccessful factor attempts are preserved; proposals do not constitute the proof. One initially redundant commutator deletion was correctly accepted and therefore failed its intended negative control; the final control collapses all commutators instead. No ambient subgroup enumeration was repeated. The earlier two Borel corrections in 3.15 and 3.16 remain in the acquired 350-entry packet. Full parabolic and quotient structures, other source claims and whole-source registration are separate obligations.

The 312 distinguished Higman–Sims parabolics

Human Compression project

This component concerns the actual distinguished parabolic subgroups in the 86 retained Higman–Sims models and the corresponding 312 node labels in the fixed supplement. The source-to-model type permutation, exact subgroup orders and actual generators are acquired from the accepted local-data and source-coordinate components. The 398 Borel/kernel occurrence maps are also acquired. The present task is the full parabolic and its extension over the *displayed* kernel. Neither a group order nor an unproved finite presentation is used as a naming certificate.

Theorem 1. *For all 312 source/model slots, the supplied literal certificates identify the actual parabolic, its distinguished kernel and quotient, with the splitting and direct-product assertions recorded in the supplied source ledger. The 61 raw source label strings and all parentheses and kernel boxes are retained. Exactly 26 slots require a correction, including five slots whose kernel correction was already acquired. The other labels are verified or made more precise. These are statements about the displayed subgroups, not a classification of all parabolic conjugacy classes or geometries.*

The finite statements in this theorem require the complete verification interface described below. This working manuscript carries no acceptance or whole-source credit before that interface and its distinct review close.

Literal transport

Lemma 1. *Let $M = \langle g_1, \dots, g_s \rangle$ be a finite permutation model whose complete right-generator Cayley graph is known. Suppose permutations $h_1, \dots, h_s$ define images y_m along the discovery words, with $y_1 = 1$, and every edge satisfies $y_{mg_i} = y_m h_i$. Then $m \mapsto y_m$ is a homomorphism. If its values are distinct, contain every generator of an actual subgroup P, and $|P| = |M|$, it is an isomorphism onto P.*

Proof. Induction on a positive word gives the claimed image of every product. Every element of a finite generated group has such a word, including inverses. Thus all products are respected. Distinctness proves injectivity; containment and the independently acquired order give equality with P. □

The enumerator closes from the identity under every generator and checks its resulting size; the expected size never truncates an open traversal. All non-discovery edges are checked. Hence a GAP isomorphism proposal is only a way of choosing the h_i. The proof uses the literal permutations and complete edges, with no group-recognition or presentation assumption.

For an extension let K be the actual listed kernel, and let M be the named quotient model. Enumerate K, verify its order, and check that every h_i normalizes it. The discovery lifts y_m must satisfy

$$y_m h_i y_{mg_i}^{-1} \in K.$$

The checked normalization makes coset multiplication well defined, so the same word argument gives a homomorphism $M \to \langle K, h_i \rangle / K$. Requiring all ky_m to be distinct proves injectivity and the exact order $|K||M|$. Every actual generator of P must occur among them. The acquired order $|P| = |K||M|$ then proves equality with P, including the actual kernel embedding. If all edges hold exactly, the y_m form a section. If the lifts also commute with every kernel generator, this is an internal direct product. A quotient certificate alone asserts neither splitting nor nonsplitting.

Complete section tests

Once the quotient map and ordered generators of M are fixed, a section must send g_i to one of $k_i h_i$, $k_i \in K$. Enumerating every ordered lift tuple and testing every Cayley edge therefore covers every section. An exact edge solution is automatically a right inverse of the quotient map and hence injective. Conversely every section supplies such a solution. The cap on the number of tuples rejects oversized tests; it never reports nonsplitting after a partial enumeration.

There are 4096 lift tuples for each of the three $|K| = 64$, $M = S_5$ cases, and all fail. Thus the three printed colons in these cases are false. The four extra central C_2 models used to identify unboxed full parabolics have respectively $4, 8, 8, 8$ lift tuples and $2, 4, 4, 4$ sections. These central subgroups are additional witnesses and are not substituted for the source's boxed kernels. A separate inner E_2^4/S_5 test has 256 tuples and 32 sections. Its purpose is a precise reassociation of one actual named model; it does not identify arbitrary groups sharing the same extension notation.

Names and parenthesized factors

The acquired named-model library gives 244 full-parabolic maps. Six further full groups are identified as $A_5 \times (5:4)$ by explicit disjoint actions; four are $C_2 \times M_{10}$ or $C_2 \times \mathrm{Aut}(A_6)$ through literal central sections. The remaining 58 are specified by their actual kernel/quotient extension certificates and fixed permutation actions. All 108 nontrivial boxed kernels have quotient certificates. There are 56 section certificates for these exact source kernels and 29 centralizing sections in total, including the four additional central witnesses. No failure of a discovery procedure is used as a nonsplitting assertion.

The elementary naming models are explicit: $\mathrm{GL}(3,2)$ on the seven nonzero binary vectors is $L_3(2)$; adjoining translations gives $\mathrm{E}_2^3 : L_3(2)$. Affine transformations of $\mathbb{F}_5$ give $5:4$. Signed coordinate matrices on $\mathbb{F}_3^2$ give $3^2 : D_8$. The matching-sign subgroup of $S_3 \times S_4$, over the normal Klein four in the second factor, is $2^2 : (3^2 : 2)$. The full product is $2^2 : (3^2 : 2^2)$. Finally $\mathbb{F}_9 = \mathbb{F}_3[u]/(u^2+1)$ has a primitive element $1+u$ of order8; Frobenius acts on it by cubing. Translations, multiplication and Frobenius therefore give $3^2 : (8:2) = \mathrm{A\Gamma L}(1,9)$. Every use of these names has its own complete literal transport.

Nested accepted trees are not reassociated by notation alone. The two actual full models named $(2^4 : A_5) : 2$ have separately checked normal E_2^4 and S_5 complements. Both actual 576-order models needed for $2^4 : (3^2 : 2^2)$ have checked E_2^4 and $S_3 \times S_3$ complements. One of these is additionally identified as $S_4 \times S_4$. The quotient of slot $66 : P3$ has its own checked $\mathrm{E}_2^4 : S_5$ section. Model IDs here are local to their exact bound input files, never an order-based alias. For $A_5 : D_{12}$ the transported full model is $S_5 \times S_3$: $A_5 \lhd S_5$ has a transposition complement and $C_2 \times S_3 = D_{12}$. For $([2] \times A_6).2^2$, the stronger actual product $C_2 \times \mathrm{Aut}(A_6)$ supplies the normal intermediate $C_2 \times A_6$.

Source corrections

The full slot table records source label, physical page, source type, model type, the untouched raw expression and box span, and the exact certificate for every assertion. The following descriptions explain all new correction patterns; the five acquired kernel corrections remain explicitly attributed to their prior components.

- The three $[4.2^4] : S_5$ extensions do not split over the displayed kernel, by complete lift exhaustion.
- The ten quotients printed $2^3 \times L_3(2)$ are $2^3 : L_3(2)$. The latter affine group has trivial centre: commuting with translations forces a central element's linear part to be the identity, and a translation fixed by every linear map is zero.
- The quotient printed $2 \times D_{10}$ is $5:4$; an element of order4 distinguishes these groups. Quotients printed 2 in source3.1 and3.20 have order4 and are 2^2. Source3.4 type2 has quotient D_{12}, not S_3, in addition to its already corrected kernel; its corrected exact sequence makes no splitting assertion.
- The two unboxed $2 \times A_6$ labels have actual order1440, with groups $C_2 \times M_{10}$ and $\mathrm{Aut}(A_6)$ respectively. Source3.24 has full group $\mathrm{Aut}(A_6)$ with trivial centre. Even if its label means $(C_2 \times A_6).C_2$, the characteristic C_2 centre of that normal index-two subgroup is fixed by every conjugation and remains central in the whole group. Thus both conventional readings are excluded.
- Source4.24 type1 is $U_3(5) : 2$ and source4.28 type1 is $U_3(5)$, with orders252000 and126000 respectively. Literal named transport, not just the corrected type sizes, establishes these identifications.

The unparenthesized source4.12 type3 expression is read sequentially as $(K \times \mathrm{E}_2^2) : (\mathrm{E}_3^2 : \mathrm{E}_2^2)$. This is verified: its actual boxed K of order4 lies in a checked normal elementary group of order16, has an internal elementary complement there, and the normal group has a checked $S_3 \times S_3$ complement. The full group is $S_4 \times S_4$. Noncentrality of K in the whole group does not refute this intermediate direct-product assertion. The earlier working interpretation is retained as a reader correction, not a source error.

The generic dot is used only for an extension; it is not a nonsplitting claim. The exact permutation model and the maps disambiguate extensions when a conventional expression alone does not. No parabolic conjugacy, residual primitivity, geometry enumeration, automorphism-group or complete original recovery is inferred by this component.

Verification boundary

The final checker must bind the exact source/type join, actual generator lists and acquired orders, each named-model proof, and all proposal maps before rebuilding and checking every used literal edge/coset certificate. The small controls compare with independent full-multiplication or homomorphism-graph oracles, in ordinary and sanitized builds. Damaged models, maps and interfaces must reject. Development failures are retained: an omitted generator in one elementary constructor, an initial map filename collision, and a checker that initially expected an unavailable proposal. None supplies mathematical evidence. Whole-source and classification credit remain zero until the separate source-wide synthesis.

Faithful point shadows and automorphisms of twenty-four Higman–Sims geometries

L091 finite classification component

Let $G = \mathrm{HS}$, of order 44 352 000, in the acquired faithful permutation action on $\Omega = \{1, \ldots, 100\}$. Let N be the acquired permutation group inducing the complete $\mathrm{Aut}(G)$, of order 88 704 000, with $C_N(G) = 1$. We use the fixed representatives in `full_fusion03/result.g`, their already established geometry and flag-transitivity properties, and their actual parabolic subgroups $P_1, \ldots, P_r$. Identifiers below refer to that file, not to historical diagram numbers. Automorphisms preserve every type; correlations may permute types. Put $B = \bigcap_i P_i$ and $A = \mathrm{Aut}(G)$.

Theorem 1. *Under the acquired finite-group inputs and the exact graph-automorphism premise stated below, the full groups of the following geometries are:*

Representatives	$\mathrm{Aut}(\Gamma)$	$\mathrm{Cor}(\Gamma)$
$19, 20, 41, 44, 47, 79, 80$	G	G
67	G	A
$83, 84$	G	$G \times C_2$
$6, 30, 31, 40, 75, 76$	A	A
$28, 60, 72, 77, 81, 82, 85, 86$	A	$A \times C_2$

These are all twenty-four representatives having a type of size 100*. The assertion is about full groups, not only normalizers of the specified G action.*

Restriction to point shadows

For each type a of size 100, the actual subgroup P_a fixes exactly one point $u_a \in \Omega$. Its order equals $|G|/100$, so it is exactly G_{u_a}. Thus $P_a g \mapsto u_a^g$ is a bijection from that type to Ω. For the base object of type j, its shadow on type a is $u_a^{P_j}$. This follows directly from right-coset incidence: the incident objects of type a are $P_a p$ for $p \in P_j$.

Lemma 2. *Let a transitive type have base stabilizer P. An equivariant encoding of its objects is injective if and only if the stabilizer in G of its base encoding is P. Once certain types are pointwise fixed by an automorphism, so are all objects encoded injectively by their shadows or by their sets of neighbours in those types.*

Proof. An equivariant encoding is the map from $P\backslash G$ to the orbit of its base value. Its fibres have size $[G_{\mathrm{value}} : P]$. An automorphism fixing the encoded types fixes every such set or tuple of their objects, and therefore every object having a unique encoding. This also proves the iterative assertion. □

The literal calculation checks that every 100-object type is determined by its shadow on any other such type, including itself. For all other types start with the ordered tuple $(u_a^{P_j})_a$. Its stabilizer is the intersection of the set stabilizers. Fifteen families already have every object determined this way. In the others, adjoin sets of encoded neighbours from already identified types. If type k is identified, the neighbours of the base type-j object are precisely the orbit

of its base type-k encoding under P_j. The program checks the orbit length $[P_j : P_j \cap P_k]$, then intersects the current encoding stabilizer with its set stabilizer. Only a type whose resulting stabilizer equals its actual P_j is admitted for use in later steps. There are 98 types in total. All are identified. Representative 40 needs two successive neighbour steps; no unproved direct-shadow injectivity is used there. Lemma 2 proves faithful restriction of the full typed automorphism group to each of the 38 possible 100-object anchors.

The intrinsic graph bound

The orbits of a point stabilizer on Ω have sizes $1, 22, 77$. Transport its orbit of size 22 to obtain a simple undirected graph Δ of degree 22. The two unordered pair orbits have sizes 1100 and 3850. These assertions are checked in the actual action. For a type-j shadow $S = u_a^{P_j}$ let e be the number of edges of Δ inside S, and put $f = \binom{|S|}{2} - e$. Double counting pairs and incident objects gives the two common-incidence counts

$$\frac{[G : P_j]e}{1100}, \qquad \frac{[G : P_j]f}{3850}. \tag{1}$$

This formula counts objects, so repeated point shadows cause no error. For every anchor at least one type has unequal counts in (1). Consequently the vector of common-incidence counts over all types partitions the distinct point pairs into exactly these two orbits. An arbitrary correlation permutes the coordinates of this vector and maps an anchor to another 100-object type. The two pair relations have different valencies, so the degree-22 relation is preserved. Thus every correlation induces an isomorphism of the corresponding copies of Δ.

The complete nauty computation for this single 100-vertex graph gives order 88 704 000. Every returned generator and every acquired generator of N is checked on all adjacencies. Hence the full graph group is exactly N. The literal permutation calculation also checks $N' = G$; in particular G is characteristic in the graph group. For any correlation c and any $g \in G$, its induced point action therefore makes cgc^{-1} equal on the target anchor to some $h \in G$. The typed automorphism $h^{-1}cgc^{-1}$ is trivial on that anchor. Faithfulness from Lemma 2 makes it trivial everywhere. We have proved that every full correlation normalizes the specified G.

Exact lifts and the group structures

Write S and T for the ordered and setwise stabilizers, respectively, of $(P_1, \dots, P_r)$ in the complete group $N = \mathrm{Aut}(G)$. An element $t \in T$ with $P_i^t = P_{\pi(i)}$ acts by

$$P_i g \longmapsto P_{\pi(i)} g^t. \tag{2}$$

This is well defined and preserves incidence in both directions. It fixes the base chamber setwise. Conversely, a correlation normalizing G can first be moved by G so that it fixes that chamber. Its induced automorphism of G then belongs to T; faithfulness of the action and the coset description give (2). The ordered analogue gives S. Consequently

$$|\mathrm{Aut}(\Gamma)| = |G : B||S|, \qquad |\mathrm{Cor}(\Gamma)| = |G : B||T|. \tag{3}$$

The complete type-lift calculation tests every type permutation, using subgroup orders and orders of pairwise intersections only to reject impossible cases. For each remaining permutation it successively transports actual subgroups in N, replacing the transporter group by the stabilizer of each target already placed. The transporter-coset induction proves that this loses no solution. Every positive witness is checked on the whole ordered family. The group generated by S and these witnesses has order $|S|$ times the number of successful type permutations, and stabilizes the family. It is therefore exactly T.

For completeness, let $C = C_{\mathrm{Cor}(\Gamma)}(G)$. Centralizing correlations are the maps obtained from inner elements $t \in T \cap G$ in (2), followed by the inverse right action of t. Their kernel is B, and all centralizing correlations arise this way by moving the image chamber back. Thus $C \cong (T \cap G)/B$ (an opposite-group convention gives the same isomorphism). Direct calculations give $S \cap G = B$, so C meets the typed automorphism group trivially. The quotient above is always trivial or of order two; its actual quotient generators are retained. The conjugation image of $\mathrm{Cor}(\Gamma)$ in $\mathrm{Aut}(G)$ has order $|\mathrm{Cor}(\Gamma)|/|C|$, hence is G or A.

If $C = 1$, this identifies the full group with that image. If $|C| = 2$, it is central in the full group, and in every such row the typed automorphism group has order $|\mathrm{Cor}(\Gamma)|/2$. Their trivial intersection and full product order give the direct product asserted in Theorem 1. In particular the factor C_2 is not inferred merely from a doubled cardinality. The exceptional row 67 has typed group G, trivial C, and full conjugation image A.

Inputs, attribution and remaining scope. The acquired complete HS action, family enumeration, and its exact Aut/type quotient are explicit premises. GAP's finite permutation and subgroup algorithms and nauty's complete graph-automorphism algorithm are retained computational inputs. The scripts retain the literal shadow encodings, positive type lifts and every rejected type permutation; no subgroup or geometry census is repeated. Small literal incidence controls test restriction, iterative shadows, type interchanges and a nonfaithful duplicate-shadow example. The shadow and transporter arguments are elementary and overlap the accepted common incidence theory. The historical L096 naming treatment retained the source's full typed-group assertion; it is not an upper-bound premise here. This component supplies a new check of the required bounds, not a novelty claim for the original results.

Representatives $14, 15, 35, 36$ have separate acquired full-group proofs. The remaining 58 representatives are outside the present result. All 86 source-diagram identifications and the global source synthesis remain separate. The source is D. Leemans, *The residually weakly primitive geometries of HS*, Australasian Journal of Combinatorics 33 (2005), 231–236, together with the explicitly retained 6 October 2003 supplementary diagrams. No whole-source, measurement, formalization or new-question credit is claimed by this component.

Higman–Sims groups from intrinsic collections of small types

Private finite application

Let $G = \mathrm{HS}$ in the acquired faithful action of degree 100. The complete coset catalogue and the identified automorphism group $N = \mathrm{Aut}(G)$, of order 88 704 000, are retained inputs. We use the source coordinates of the archived supplement to D. Leemans, *The residually weakly primitive geometries of HS*, Australas. J. Combin. **33** (2005), 231–236. The claims below concern the explicitly listed families only.

Restriction and faithfulness

Select all types having cardinality in a specified finite set. The collection is intrinsic, including all repeated cardinalities. Its induced incidence graph has one additional marker per type. An object is coloured by its type cardinality; marker colours are distinct from object colours and otherwise also depend only on cardinality. A marker is joined to exactly the objects of its type. Thus the colour-preserving graph group is exactly the correlation group of the selected incidence system, with no artificial distinction between types of the same size.

Lemma 1. *Suppose the image E of G on the selected objects is faithful and is characteristic in their full induced graph group. Suppose every typed automorphism fixing the selected objects fixes all objects of the original geometry. Then every correlation of the original geometry normalizes its specified G action.*

Proof. A correlation c restricts to the intrinsic selected system. For every $g \in G$, characteristicity gives $h \in G$ whose restriction equals that of cgc^{-1}. The typed automorphism $h^{-1}cgc^{-1}$ fixes every selected object, so is trivial by hypothesis. Consequently $cgc^{-1} = h \in G$. □

Here faithfulness is checked through the actual coset action and successive incidence descriptions. Write P_i for the base parabolics and b_j for the selected base objects. The first description of a type-i object is its tuple of shadows on the selected types. At the base object this is $(b_j^{P_i})_j$. Its stabilizer in G is computed as an intersection of actual set stabilizers. It equals P_i exactly when this description is injective within type i.

If some types are not yet distinguished, use a type k whose description has already been proved injective. Append to the description of type i the set of descriptions of its type-k neighbours. The base set is the orbit of the base description of type k under P_i; its size is checked to equal $[P_i : P_i \cap P_k]$. Intersecting the old stabilizer with this new set stabilizer gives the refined stabilizer. Each step uses only a previously established type. Induction proves that a typed automorphism fixing the selected objects fixes every described object. When all final stabilizers are P_i, the hypothesis of the lemma follows. This is the already acquired iterative-shadow argument, applied to joint selected actions.

Literal finite identifications

For each selected type the saved ordered generator images define an actual homomorphism from the acquired G to the joint permutation group E. The verifier checks that it is bijective. The base vertex has stabilizer equal to the image of P_i, and its orbit is the entire type. Hence $P_i g \mapsto b_i^g$ is the required coset bijection; no agreement of historical vertex labels is assumed.

For each ordered pair of selected types, the neighbours of the base vertex are checked to be exactly the appropriate parabolic orbit. Every generator preserves every coloured adjacency row. Equivariance then gives all incidences and nonincidences. The complete nauty group is a computational premise; every returned generator is also checked on every coloured row. GAP verifies the exact order of the generated group D, containment $E \leq D$, and $D' = E$. This last equality establishes the characteristic subgroup needed in the lemma, rather than inferring it from a numerical order alone.

The shadow verifier reconstructs all initial orbits and stabilizers, then every recorded refinement in its actual chronological order. It checks the exact finite descriptions, the complete neighbour orbit, the saved stabilizer generators, and equality to every parabolic. In all nine applications below, the initial shadows already distinguish every type: there are no refinement steps. Successful candidate output or agreement with a source order is not substituted for these checks.

From normalization to full groups

Let $B = \bigcap_i P_i$, and let S, T be respectively the ordered and setwise family stabilizers in the complete $N = \mathrm{Aut}(G)$. If $P_i^t = P_{\pi(i)}$, the rule $P_i g \mapsto P_{\pi(i)} g^t$ is an incidence isomorphism. Conversely a correlation normalizing G can be moved by G to fix the base chamber and then has precisely this form. Therefore

$$|\mathrm{Aut}(\Gamma)| = |G : B||S|, \qquad |\mathrm{Cor}(\Gamma)| = |G : B||T|.$$

All type permutations are tested. Orders and intersection orders reject only impossible permutations; successive complete subgroup transporters handle the rest, and every positive lift is checked on the whole family. This is the acquired normalizer argument, with literal new families.

The computation also checks $S \cap G = B$ and the actual quotient $(T \cap G)/B$. The latter is the centralizer of the specified G in the full correlation group. When it is trivial, the full group identifies with its conjugation image in N. When it has order two, it is central. If the typed group is G and the full group has twice its order, the central factor has trivial intersection with G and their product has full order. This proves $G \times C_2$, not merely an order equality.

Theorem 2. *For the nine acquired families in the table, the full type-preserving automorphism group is G. Their full correlation groups are as shown. Here v counts selected objects and type markers, $d = |D|/|G|$, and $t = |T|/|B|$; the source label uses the acquired exact coordinate map.*

Family	Source	v	d	$\lvert B\rvert = \lvert S\rvert$	t	$\mathrm{Cor}(\Gamma)$
24	3.12	1454	1	72	1	G
25	3.15	2555	2	72	2	$G \times C_2$
38	4.28	1454	1	4	1	G
57	4.19	706	4	8	2	$G \times C_2$
58	4.17	1454	1	8	1	G
59	4.16	1807	2	8	2	$G \times C_2$
63	4.22	1807	2	8	1	G
64	4.21	1454	1	8	1	G
65	4.20	1454	1	8	1	G

Proof. For each row the literal checks establish $D' = E \cong G$, injectivity of every initial shadow, and the complete ordered and setwise family stabilizers. The preceding restriction lemma and normalizer formulas therefore give the full groups. When $t = 2$, the verified centralizer $(T \cap G)/B$ has order two, so the direct-product argument applies. The larger selected graph group for family 57 is only an upper-bound vehicle: the full geometry has $t = 2$, not $d = 4$. □

Finite scope. Families 21, 22, 37, 39, 54 and 56 are explicitly left open by this component; their individual 60-second model attempts were interrupted. Candidate attempts that reached their individual deadlines remain unproved cases. In those cases the recorded normalizer groups are only lower bounds. No ambient subgroup or geometry enumeration is repeated; source names, numerical classification and software completeness remain explicit premises. No whole-source completion, new conjecture, formalization or measurement claim is made.

Six full geometry groups from an intrinsic 352-object relation

Let $G = \mathrm{HS}$ in the acquired faithful permutation model. We retain the complete coset catalogue, the literal full $\mathrm{Aut}(G)$ action, and the reviewed source coordinates for D. Leemans, *The residually weakly primitive geometries of HS*, Australasian Journal of Combinatorics 33 (2005), 231–236, with its archived supplementary diagrams.

Theorem 1. *For the following six families the full type-preserving automorphism group is G. Their full correlation groups, allowing type permutations, are as follows. Here B is the base chamber stabilizer, and S, T are the ordered and setwise parabolic-family stabilizers in* $\mathrm{Aut}(G)$.

Family	*Source*	$\lvert B\rvert$	$\lvert S\rvert$	$\lvert T\rvert$	$\mathrm{Cor}(\Gamma)$
21	*3.13*	*72*	*72*	*72*	G
22	*3.16*	*72*	*72*	*144*	$G \times C_2$
37	*4.31*	*4*	*4*	*8*	$G \times C_2$
39	*4.29*	*4*	*4*	*4*	G
54	*4.14*	*8*	*8*	*8*	G
56	*4.18*	*8*	*8*	*16*	$G \times C_2$

The proof uses explicit finite maps and exact group operations, with the complete coloured-graph algorithm as a stated computational premise.

A smaller intrinsic structure. Every listed family has exactly one type a of cardinality $n = 352$. It is therefore fixed setwise by every correlation. Let b be its base object, $P_a = G_b$, and let O_k be the P_a-orbits on that type, of sizes d_k. For each type j, the point shadow of its base object is $S_j = b^{P_j}$. If e_{jk} counts the ordered distinct pairs in S_j belonging to pair orbit k, then the number of type-j objects incident with both entries of such a pair is

$$m_{jk} = \frac{[G : P_j]e_{jk}}{352d_k}.$$

To prove this, count pairs consisting of an object and an ordered pair in its shadow. The total is $[G : P_j]e_{jk}$; the pair orbit has $352d_k$ elements. Transitivity makes the fibre size constant. Objects with equal shadows are still counted with their proper multiplicity. For $j = a$, equality is the within-type shadow convention, so all off-diagonal counts vanish.

The sorted multiset of $([G : P_j], m_{jk})$ over all types is invariant under every correlation. Its distinct values therefore define intrinsic relations on ordered distinct points. In all six literal models the point-stabilizer subdegrees are $1, 175, 175, 1$; the non-diagonal signatures define two relations of valencies 176 and 175. The unique valency 175 relation is checked to be symmetric. It gives a graph on just 352 vertices. This construction never requires a transversal or permutation action on any of the larger object types.

Literal coordinates and faithfulness. The recorded generator images give a bijective homomorphism from the actual acquired G to a permutation group E on 352 points. The base-point stabilizer is checked to be the actual image of P_a. All saved elements $t_x \in G$ send the base point to the corresponding point x. These checks establish the coset identification $P_a g \mapsto b^g$,

without any assumption about historical labels. For a pair (x, y), its orbital is determined by the E_b-orbit of $y^{t_x^{-1}}$. This computes every e_{jk} from the saved actual shadows and reconstructs every graph row by equivariance.

A point shadow determines a type-j object exactly when its setwise stabilizer in E is the image of P_j. All types in families $21, 22, 37, 54, 56$ meet this condition directly. In family 39, the first type initially has shadow stabilizer of order 1440, whereas its parabolic has order 32. The other three types are already identified. Adjoin the set of encodings of its type-3 neighbours. This is the actual orbit of the base type-3 encoding under P_1; its size is $[P_1 : P_1 \cap P_3] = 2$. Its stabilizer inside the old stabilizer has order 32 and equals the actual image of P_1. The saved certificate also records the preceding redundant refinement by type 2, which leaves order 1440; both entries are checked in dependency order, including their complete neighbour orbits.

Thus any typed automorphism fixing the 352 objects pointwise fixes every object. This is the acquired injective-encoding lemma: the base encoding has stabilizer exactly its parabolic, and induction through previously identified types gives the same implication for all successive neighbour descriptions.

Characteristic group and full correlations. A complete nauty calculation gives graph-group order 88 704 000 in every case. Every returned generator is checked on every adjacency row. For their literal permutation group D, GAP checks

$$|D| = 88\,704\,000, \qquad E \leq D, \qquad D' = E.$$

In particular E is characteristic in the full graph group. If c is any correlation and $g \in G$, the action of cgc^{-1} on the intrinsic point type therefore agrees with some $h \in G$. The typed automorphism $h^{-1}cgc^{-1}$ fixes that type pointwise, so the preceding faithfulness argument makes it trivial. Hence all full correlations normalize the specified G.

The acquired chamber-stabilizer argument now gives

$$|\operatorname{Aut}(\Gamma)| = |G : B||S|, \qquad |\operatorname{Cor}(\Gamma)| = |G : B||T|.$$

Complete successive subgroup transporters test every possible type permutation. Every positive witness is checked on the entire actual parabolic family; impossible order or intersection-order patterns are only rejection criteria. The resulting S, T are those in the table, with $S \cap G = B$. Therefore every typed automorphism lies in G. The centralizer of G in the full correlation group is the checked quotient $(T \cap G)/B$. It is trivial for $21, 39, 54$, and has order two for $22, 37, 56$. In the latter cases it is central, intersects the centreless G trivially, and together with G has the full correlation order. This proves the direct products in the theorem.

Attribution and boundary. The common-incidence formula, shadow lemma and normalizer argument are acquired theory; this is their finite application at degree 352. Exact GAP arithmetic and nauty completeness remain computational premises. The preceding failed joint-action attempts are preserved; this changed reduction repeats no ambient subgroup or geometry census. The source-coordinate map is retained separately from structural names. No abstract Borel-name identification, whole-source completion, historical novelty, question resolution, Lean or measurement claim is supplied by this component.

Automorphism groups from the link of a point

Finite application to six Higman–Sims geometries

Let $G = \mathrm{HS}$, with its acquired faithful permutation model, complete coset catalogue and identified group $N = \mathrm{Aut}(G)$. We retain the source coordinates of the supplement to D. Leemans, *The residually weakly primitive geometries of HS*, Australas. J. Combin. **33** (2005), 231–236. Our purpose is to determine the full groups in six specified geometries. The numerical classification, source coordinates and finite group interfaces are acquired premises. The reduction to a point link is proved here.

Lemma 1 (A point link bound)**.** *Let $G \le A \le \mathrm{Sym}(\Omega)$, where G is transitive on the finite set Ω. Suppose A preserves a family $\mathcal{B}$ of distinct subsets of Ω. For $b \in \Omega$, put*

$$\mathcal{L}_b = \{B \setminus \{b\} : B \in \mathcal{B},\ b \in B\}.$$

Let L be the group of permutations of $\Omega \setminus \{b\}$ preserving $\mathcal{L}_b$. Then $|A| \le |\Omega||L|$. In particular, if $|\Omega||L| = |G|$, then $A = G$.

Proof. Every element of A_b permutes the blocks containing b and therefore preserves their deletions. Its restriction to $\Omega \setminus \{b\}$ determines it, so A_b embeds in L. Since G is transitive, so is A, and orbit–stabilizer gives $|A| = |\Omega||A_b| \le |\Omega||L|$. The last assertion uses the actual containment $G \le A$. □

We compute L as the complete group of the bipartite incidence graph whose two parts are the points and the distinct members of $\mathcal{L}_b$. The parts have different colours. Distinct members have distinct point neighbourhoods, so an automorphism fixing every point fixes every block. Conversely a permutation preserving the family extends uniquely to this graph. Thus its full coloured graph group is exactly L; there is no unaccounted permutation kernel on repeated blocks.

Blocks determined by incidence. Choose a type a of cardinality 176, and let $\Omega = \{P_a g : g \in G\}$. For each other type j, take the distinct point shadows of its objects. At the base object the shadow is $S_j = b^{P_j}$, so these shadows form the orbit S_j^G and their number is $[G : G_{S_j}]$. Among types with $2 < |S_j| < 174$, choose the least triple

$$\big([G : G_{S_j}],\ [G : P_j],\ |S_j|\big)$$

in lexicographic order. Take the union of the shadow families of *all* types attaining this triple, discarding repeated subsets. This gives $\mathcal{B}$. Every type-preserving automorphism preserves it: the construction uses only incidences, cardinalities and distinct shadows. The group notation supplies a way to compute these intrinsic quantities.

Faithfulness of the restriction. The acquired shadow lemma supplies a sufficient condition for the full typed group to act faithfully on Ω. A type-j object's initial description is its point shadow. If its stabilizer in G equals P_j, the description is injective within that type. For a type not yet identified, append the set of descriptions of its neighbours of an already identified type k. At the base object this set is the P_j-orbit of the base description of type k, of size $[P_j : P_j \cap P_k]$. Intersecting the previous stabilizer with the stabilizer of this set gives the refined

stabilizer. Induction on these dependencies proves that a typed automorphism fixing Ω fixes every identified type. In all applications every final stabilizer equals the corresponding P_j. Only family 53 requires a refinement: type 1 uses its two type-2 neighbours.

Exact finite interfaces. For every anchor the saved images of the ordered generators give a checked bijection $\phi : G \longrightarrow E \leq \mathrm{Sym}(176)$. The verifier checks that the base point has stabilizer $\phi(P_a)$ and that the saved representatives carry it to every numbered point. These equations identify the action with the actual coset action. All point shadows, chosen triples and description stabilizers are then computed from the actual subgroup images.

The saved block list is checked without rediscovering its orbits. Every generator permutes the whole list, every selected base shadow is present, and a traversal from these seeds along generator edges reaches every listed block. Closure gives one inclusion in the required union; reachability gives the other. Since the action is finite, positive generator words suffice. This proves exact equality with $\mathcal{B}$. The link is formed directly from this verified list. Complete nauty output is an explicit computational premise. Each returned generator is also checked on every link block and on its unique extension to the graph.

Theorem 2. *In the six families below, the full type-preserving group equals the specified G. The full correlation groups are as shown. For each listed anchor the complete link group has order* 252 000*.*

Family	Source	Anchor	$(\lvert\mathcal{B}\rvert, \lvert B\rvert)$	$\mathrm{Cor}(\Gamma)$	$\lvert\mathcal{L}_b\rvert$
52	4.26	3,4	$(30800, 6)$	$G \times C_2^2$	1050
53	4.25	3,4	$(30800, 6)$	$G \times C_2$	1050
66	4.12	1	$(5775, 16)$	G	525
68	4.9	3	$(15400, 12)$	G	1050
		4	$(36960, 6)$		1260
69	4.10	3,4	$(36960, 6)$	$G \times C_2$	1260
70	4.8	4	$(36960, 6)$	G	1260

Proof. The faithful restriction, intrinsic block family and point link bound give $|\mathrm{Aut}(\Gamma)| \leq 176 \cdot 252000 = 44352000 = |G|$. The specified G already acts by typed automorphisms; hence equality holds as groups of automorphisms. In particular G is the kernel of the type-permutation action of $\mathrm{Cor}(\Gamma)$, and is normal there.

Put $B_0 = \bigcap_i P_i$. The acquired normalizer theorem therefore applies. If S, T are the ordered and setwise family stabilizers in the complete N, it gives

$$|\mathrm{Aut}(\Gamma)| = |G : B_0||S|, \qquad |\mathrm{Cor}(\Gamma)| = |G : B_0||T|,$$

and identifies the centralizer of G with $(T \cap G)/B_0$. Every type permutation is tested by complete successive subgroup transporters, and each positive lift is checked on all parabolics. The six values of $(|B_0|, |S|, |T|)$, in table order, are

$$(8, 8, 32), \quad (8, 8, 16), \quad (16, 16, 16), \quad (20, 20, 20), \quad (20, 20, 40), \quad (20, 20, 20).$$

The centralizer has respectively order $4, 2, 1, 1, 2, 1$; in the first case every element squares to the identity, proving that it is C_2^2. As G is centreless, its intersection with this centralizer is trivial. Their product has the computed full correlation order, which proves the direct products displayed in the table. □

Scope of the verification. All ten anchors are checked, with all four types identified in each. Damaged controls test closed extra orbits, omitted or duplicated blocks, wrong seed families, wrong actual subgroup data and missing faithfulness. Ordinary and optimized Python reports must agree. The computation repeats no ambient subgroup or geometry census. This application supplies six full-group determinations; it does not complete the source manuscript, settle an open conjecture or supply a Lean formalization.

Common incidences determine eleven full geometry groups

Let $G = \mathrm{HS}$, and let $A = \mathrm{Aut}(G)$, in the acquired faithful models of orders 44 352 000 and 88 704 000. We use the reviewed classification and source coordinates for the coset geometries in D. Leemans, *The residually weakly primitive geometries of HS*, Australasian Journal of Combinatorics 33 (2005), 231–236, with the retained supplementary diagrams. The original results remain attributed to that source.

Theorem 1. *The following are the full type-preserving automorphism groups and full correlation groups, respectively. Correlations may permute types.*

Representative	*Source label*	$\mathrm{Aut}(\Gamma)$	$\mathrm{Cor}(\Gamma)$
11	*3.25*	A	A
12	*3.26*	A	$A \times C_2$
23	*3.14*	G	G
26	*3.11*	A	$A \times C_2$
27	*3.10*	A	$A \times C_2$
29	*3.7*	A	A
49	*4.32*	A	A
50	*4.33*	A	A
55	*4.15*	G	$G \times C_2$
71	*4.7*	A	$A \times C_2$
74	*4.3*	A	A

The finite proof uses exact permutation arithmetic and the complete graph automorphism algorithm specified below.

Common-incidence relations. Fix any type a of size $n = 1100$, with base object b and stabilizer P_a. The acquired group is transitive on that type. Let O_k be the orbits of P_a on its objects, of sizes d_k; omit the singleton $\{b\}$. They parametrize the orbits on ordered distinct point pairs, of respective sizes nd_k. For another type j, let $S_j = b^{P_j}$ be the point shadow of its base object. Define e_{jk} to be the number of ordered distinct pairs in S_j belonging to pair orbit k. Then the number of type-j objects incident with both members of a pair in orbit k is

$$m_{jk} = \frac{[G : P_j]e_{jk}}{nd_k}. \tag{1}$$

Indeed, count triples consisting of a type-j object and an ordered pair in its shadow. Counting first by objects gives the numerator; transitivity on the pair orbit makes the count per pair constant. This argument counts objects even when two objects have identical shadows. For $j = a$, the shadow convention is equality within that type, so all off-diagonal counts are zero.

For a distinct pair, form the sorted multiset of pairs $([G : P_j], m_{jk})$ over all types j. This signature is intrinsic: a correlation merely permutes its entries and sends a to a type of the same size. Equal signatures partition the distinct point pairs into intrinsic relations. In representatives 11 and 23 these relations have valencies 280 and 819. In each remaining representative, at every 1100-object anchor, there is exactly one relation of valency 105. Use the valency 280 relation in the first two cases and the valency 105 relation in the others. The literal checks verify symmetry, so each chosen relation is an undirected graph.

Computing the relations in actual coordinates. The saved permutations on 1100 points define a bijective homomorphism from the actual G onto its image E. The specified base point has stabilizer exactly the image of P_a. Recorded elements $t_x \in G$ send that base point to every point x, in order. These facts prove the coset identification $P_a g \mapsto b^g$; the certificate need not assume that a producer's enumeration was a correct transversal. For each ordered pair (x, y) in S_j, its orbit index is that of $y^{t_x^{-1}}$ under E_b. Thus finite counts of these indices compute e_{jk} and hence (1). Transporting the selected base neighbour set by the t_x gives every adjacency row. All specified generators are checked on the entire graph.

Faithfulness of point restriction. We use the elementary shadow lemma from the reviewed common theory: an equivariant encoding of $P_j\backslash G$ is injective precisely when its base value has stabilizer P_j. Begin by encoding an object by its point shadow. If an already identified type k is needed, adjoin the set of encodings of its type-k neighbours. At the base object of type j this is exactly the P_j-orbit of the base type-k encoding. Its cardinality is checked to be $[P_j : P_j \cap P_k]$.

The positive verifier reads the saved encodings in dependency order. It admits an added neighbour set only after type k has been proved injectively encoded, verifies that entire orbit, and intersects the corresponding set stabilizers. The final stabilizer must equal the actual image of P_j. There are sixteen possible 1100-object anchors in these eleven families; every type is identified at every anchor. In particular, representative 49 at its third type needs successive neighbour sets to identify types 1 and 2. No direct-shadow assertion is substituted for these iterations. An automorphism fixing the point type pointwise therefore fixes every object, by induction through the same dependencies. Restriction of the full typed automorphism group to any one anchor is faithful.

The complete upper bound. For each of the sixteen graphs, a complete nauty computation returns order 88 704 000. Every returned generator is checked on every adjacency row. In the literal permutation group D they generate, the verifier checks $|D| = 88\,704\,000$, $E \leq D$ and $D' = E$. Thus E is characteristic in the full graph group. This last equality, rather than agreement of orders alone, supplies the needed invariant copy of the specified group.

Any correlation sends an anchor graph to the graph at its target anchor, by the intrinsic common-incidence definition. If c is such a correlation and $g \in G$, characteristicity implies that the action of cgc^{-1} on the target anchor equals that of some $h \in G$. The typed automorphism $h^{-1}cgc^{-1}$ fixes that anchor pointwise. Faithfulness proves it is the identity. Consequently every full correlation normalizes the specified G.

Exact normalizers and structures. Put $B = \bigcap_i P_i$, and let S, T be the ordered and set-wise stabilizers of the parabolic family in the acquired $\mathrm{Aut}(G)$. The reviewed coset normalizer theorem now applies to the full groups:

$$|\operatorname{Aut}(\Gamma)| = |G : B||S|, \qquad |\operatorname{Cor}(\Gamma)| = |G : B||T|.$$

The same exhaustive transporter calculation tests every type permutation. It verifies actual family images, not just subgroup orders. For the representatives in table order the triples $(|B|, |S|, |T|)$ are

$$(16, 32, 32),\ (16, 32, 64),\ (72, 72, 72),\ (192, 384, 768),$$
$$(192, 384, 768),\ (192, 384, 384),\ (4, 8, 8),\ (4, 8, 8),$$
$$(8, 8, 16),\ (32, 64, 128),\ (64, 128, 128).$$

In every row $S \cap G = B$. The centralizer of G in the full correlation group is the actual quotient $(T \cap G)/B$. Its order is two exactly for representatives $12, 26, 27, 55, 71$, and is otherwise one. In the former cases it is central and meets the typed group trivially; the typed group and this centralizer have full product order. Hence the displayed factors C_2 are direct factors. The

faithful conjugation action identifies the remaining factor with G or $\mathrm{Aut}(G)$ as indicated. This proves all group structures.

Attribution and verification boundary. The coset, shadow and normalizer arguments are reused from the common theory. Formula (1) generalizes the previously reviewed 100-point count; sorted signatures allow arbitrary type interchanges. The present contribution is this application and its explicit finite identifications. The original classification, exact source-coordinate joins, GAP's finite group algorithms and nauty's completeness are acquired premises. Literal small-group controls compare the formula with direct common-neighbour counts, including repeated shadows. No new classification census, historical novelty, conjecture resolution, Lean verification or whole-manuscript completion is asserted here.

Groups determined by small incidence systems

Let $G = \mathrm{HS}$ in the acquired faithful action of degree 100. We retain its complete automorphism group N, of order 88 704 000, and the accepted list of coset geometries. Representatives below refer to that list. The source labels refer to the supplementary diagrams accompanying Leemans, *The residually weakly primitive geometries of HS*, Australasian Journal of Combinatorics 33 (2005), 231–236. Their identification with the acquired representatives is separately reviewed.

Theorem 1. *The full groups of representatives* $34, 48, 51, 61, 62$ *are as follows.*

Representative	*Source label*	$\mathrm{Aut}(\Gamma)$	$\mathrm{Cor}(\Gamma)$
34	*3.3*	G	$G \times C_2$
48	*4.34*	G	G
51	*4.27*	G	G
61	*4.24*	G	G
62	*4.23*	G	G

The assertion uses the acquired finite group model and the complete graph automorphism algorithm specified below. It is not a statement that the normalizer of a specified group action is automatically the full geometry group.

Lemma 2. *Let a flag-transitive coset geometry have a faithful specified group G, with parabolics P_i. Select all types whose cardinalities belong to a fixed set. Suppose every object is determined uniquely by the tuple of its shadows on the selected types. Suppose the induced incidence system on those types has full correlation group D, and the image of G in D is characteristic. Then every correlation of the original geometry normalizes its specified G action.*

Proof. The selected types form an intrinsic collection, so a correlation c restricts to a correlation of the induced system. For $g \in G$, characteristicity gives $h \in G$ whose restriction agrees with that of cgc^{-1}. The typed automorphism $h^{-1}cgc^{-1}$ fixes all selected objects. It fixes the shadow tuple of every remaining object and hence, by uniqueness, fixes that object. Thus $cgc^{-1} = h$. □

The argument is the same restriction argument used in the separately reviewed component for the geometries having a type of size 100. Here we select every type of size 176 or 1100. This selection, including repeated cardinalities, is checked against the actual subgroup indices. No type is selected by an arbitrary historical label.

Literal coset and shadow identifications. Write X for the union of the selected types. The certificate gives permutations on X for the ordered generators of the actual G. Their homomorphism is checked and is bijective onto its image E. For each selected type the specified base vertex has stabilizer exactly the image of P_i, and its orbit is the full type. Thus $P_i g \mapsto b_i^g$ is an actual coset bijection. For every ordered pair of selected types, the neighbours of b_i are checked to be precisely $b_j^{P_i}$. Every generator preserves every incidence and nonincidence row. Equivariance therefore proves the full incidence identification, not only agreement of local degrees.

The shadow of the base object of an arbitrary type j is the tuple $(b_i^{P_j})_i$. Its setwise tuple stabilizer in E is checked to be exactly the image of P_j, in all nineteen types of these five geometries. The orbit map then has fibres of size one. This proves the uniqueness required by the lemma, without constructing all objects of the larger types.

The complete upper bounds. The graph used for the induced incidence system has one marker for each selected type, joined to all objects of that type. Objects are coloured by their type cardinality. Markers have separate colours, also grouped by cardinality. A graph automorphism therefore permutes whole types of equal cardinality; conversely every correlation of the induced system extends uniquely to the markers. In particular, repeated types are not artificially distinguished by separate colours.

For representatives $34, 48, 51, 61, 62$ the graph sizes, including markers, are respectively $2379, 1278, 2379, 1278, 1278$. Complete nauty computations give orders $88\,704\,000$ in the first case and $44\,352\,000$ in the other four. All returned generators and all specified generators of G are checked on every coloured adjacency row. The group generated by the returned permutations has the reported order and has derived subgroup exactly E. Thus E is characteristic in each full graph group, and the lemma applies.

From normalizers to full groups. We now use the previously reviewed coset normalizer argument. Put $B = \bigcap_i P_i$; let S and T be the ordered and setwise family stabilizers in $N = \mathrm{Aut}(G)$. Once every full correlation normalizes G, flag transitivity and the base chamber give

$$|\operatorname{Aut}(\Gamma)| = |G : B||S|, \qquad |\operatorname{Cor}(\Gamma)| = |G : B||T|.$$

The calculation checks every type permutation. It discards a permutation only when subgroup orders, intersection orders, or a complete successive subgroup transporter proves it impossible. Every surviving transporter is checked on the entire family. This is exactly the already reviewed method, applied to five new literal families.

The resulting values are

Representative	$\lvert B\rvert$	$\lvert S\rvert$	$\lvert T\rvert$	$\lvert(T \cap G)/B\rvert$
34	720	720	1440	2
48	4	4	4	1
51	6	6	6	1
61	8	8	8	1
62	8	8	8	1

In every row $S \cap G = B$. Consequently the typed group is exactly the specified G. For row 34, the centralizer of G in the full correlation group is $(T \cap G)/B$, of order two. It is central and meets the typed group trivially. Their product has the full order, proving $\operatorname{Cor}(\Gamma) \cong G \times C_2$. The other four full groups already have order $|G|$ and hence equal G.

Verification and scope. The verifier checks actual stabilizers, base incidences and all graph generators; it does not repeat an ambient subgroup or geometry census.

The argument reuses elementary coset, shadow and normalizer ideas already established in the project. The new contribution here is their explicit application and the finite identifications. GAP's exact finite group algorithms and nauty's completeness are computational premises; no Lean verification, novelty, conjecture resolution or whole-manuscript completion is asserted. The other source obligations remain separate.

Common incidences in fifteen geometries of the Higman–Sims group

Let $G = \mathrm{HS}$ and $A = \mathrm{Aut}(G)$, of orders 44 352 000 and 88 704 000. We use their acquired faithful permutation models and the reviewed classification of coset geometries with specified parabolics $P_1, \ldots, P_r$. Source labels refer to the retained supplement to D. Leemans, *The residually weakly primitive geometries of HS*, Australasian Journal of Combinatorics 33 (2005), 231–236. The supplement attributes geometry 3.4 to Ivanov and Shpectorov; that attribution is retained.

Theorem 1. *The full groups are as follows. Here n is the least type size, d is the valency of the intrinsic graph used in the proof, and q is its full group order divided by $|G|$. Every type of size n is checked.*

Representative	*Source*	n	d	q	$\mathrm{Aut}(\Gamma)$	$\mathrm{Cor}(\Gamma)$
3	*2.4*	*5600*	*132*	*1*	G	A
4	*2.3*	*2200*	*105*	*4*	A	A
5	*2.2*	*4125*	*28*	*2*	A	A
7	*3.30*	*5600*	*132*	*1*	G	$G \times C_2$
8	*3.29*	*5600*	*132*	*1*	G	G
9	*3.28*	*5600*	*132*	*1*	G	G
13	*3.24*	*5600*	*495*	*1*	G	G
32	*3.4*	*3850*	*15*	*2*	A	A
33	*3.5*	*3850*	*15*	*2*	A	$A \times C_2$
42	*4.40*	*5600*	*132*	*1*	G	A
43	*4.36*	*5600*	*132*	*1*	G	G
45	*4.39*	*5600*	*132*	*1*	G	A
46	*4.41*	*5600*	*55*	*1*	G	A
73	*4.5*	*2200*	*105*	*4*	A	$A \times C_2$
78	*5.6*	*5600*	*132*	*1*	G	A

The correlation group in source diagram 4.41 is printed as $A \times C_2$; the exact calculation below gives A. The other group assertions in this table agree with the source, with an omitted correlation entry understood according to the supplement's stated convention.

Counting pairs from one point. Choose an anchor type a of size n, base point b, and stabilizer P_a. Let O_k be its orbits on points, with $d_k = |O_k|$. For another type j, its base object has point shadow $S_j = b^{P_j}$. For a non-diagonal orbit, let e_{jk} count ordered pairs in S_j belonging to the G-orbit of (b, y), where $y \in O_k$. Then

$$e_{jk} = |S_j|\,|S_j \cap O_k|, \qquad m_{jk} = \frac{[G : P_j]|S_j|\,|S_j \cap O_k|}{n d_k}. \tag{1}$$

To prove the first equality, take any $x \in S_j$. Some element of P_j sends b to x. It preserves S_j and the specified G-orbit on pairs, so the number of possible second coordinates at x equals that at b. Summing this constant over S_j proves the assertion. For the second, count triples consisting

of a type-j object and a pair in its shadow. There are $[G : P_j]e_{jk}$ triples and nd_k pairs, and transitivity makes the number of objects per pair constant. This counts distinct objects even when their shadows coincide. The type-a convention is equality within that type; its off-diagonal entries are zero.

For each pair form the sorted multiset of $([G : P_j], m_{jk})$ over all types. A correlation preserves this signature, including multiplicities. Partition the distinct pairs by signature and order the signatures lexicographically. For the valency d in the table, choose the first relation with that valency. This rule is intrinsic and is compatible with passage to another anchor of the same size. It does not require the valency to determine a unique relation: in representative 73 there are two relations of valency 105. Every selected relation is verified to be symmetric and irreflexive.

Exact point coordinates and faithful restriction. For each anchor a literal ordered list of generator images defines a bijective homomorphism $\phi : G \longrightarrow E \leq \mathrm{Sym}(n)$. The saved elements $t_x \in G$ satisfy $b^{\phi(t_x)} = x$, and the stabilizer of b in E is exactly $\phi(P_a)$. These checks prove the identification of the right coset $P_a g$ with $b^{\phi(g)}$; no correctness assumption about a producer's transversal is required. Orbits of $\phi(P_a)$ and $\phi(P_j)$ give the sets in (1). The formula constructs the base neighbour set, and the t_x transport it to every adjacency row. Every specified generator is checked on every row.

At all twenty-four anchors, the stabilizer of S_j in E is exactly $\phi(P_j)$ for every j. Thus the equivariant map from type-j objects to their shadows is injective: its fibres are the cosets of the shadow stabilizer modulo the object stabilizer. Consequently a typed automorphism fixing all anchor points fixes all objects. This proves faithfulness of restriction to each anchor. No refinement by additional types is needed for these fifteen applications.

Full graph groups and normality. A complete nauty calculation gives the order $q|G|$ in the table for each of the twenty-four graphs. The integer-order extraction is exact in the specified range, and every returned generator is checked against all adjacency rows. Let D be their literal permutation group. The verifier checks its order, $E \leq D$, and the two conditions

$$e^z \in E \quad (e \in \mathcal{E}, z \in \mathcal{D}), \qquad [z, w] \in E \quad (z, w \in \mathcal{D}),$$

where $\mathcal{E}$ and $\mathcal{D}$ are the displayed generating sets. The first condition implies $E \lhd D$: conjugation gives inclusion, hence equality by finiteness. The second makes D/E abelian. The acquired G is perfect, also checked in its original model, so $E = E' \leq D' \leq E$. Therefore $D' = E$ and E is characteristic in D.

A correlation carries each anchor graph to the graph of its image anchor, hence carries their derived groups to each other. If c is a correlation and $g \in G$, the restriction of cgc^{-1} to the image anchor equals that of some $h \in G$. The typed automorphism $h^{-1}cgc^{-1}$ fixes the anchor pointwise and is therefore trivial by faithfulness. Thus the specified G is normal in the full correlation group.

Completing the geometry groups. Put $B = \bigcap_j P_j$, and let S, T be the ordered and setwise stabilizers of the parabolic family in A. The acquired coset normalizer theorem now applies to the full geometry groups and gives

$$|\operatorname{Aut}(\Gamma)| = |G : B||S|, \qquad |\operatorname{Cor}(\Gamma)| = |G : B||T|.$$

The calculation considers every type permutation. Necessary order tests only discard impossible cases. For each remaining permutation, successive transporter calculations retain precisely the automorphisms carrying all parabolics already placed to their targets. Positive maps are checked

on the whole family. The resulting orders, in the order of the theorem, are

$$(|B|,|S|,|T|) = (660,660,1320),\ (1344,2688,2688),$$
$$(1536,3072,3072),\ (12,12,24),\ (12,12,12),\ (12,12,12),$$
$$(16,16,16),\ (256,512,512),\ (256,512,1024),$$
$$(4,4,8),\ (4,4,4),\ (4,4,8),\ (4,4,8),$$
$$(64,128,256),\ (3,3,6).$$

In every row $S \cap G = B$, so the typed group acts faithfully by conjugation on G and is identified with G or A by its order. The full centralizer of G is the actual quotient $(T \cap G)/B$. It has order two precisely for representatives 7,33,73 and order one otherwise. In those three cases the centralizer is central, meets the typed group trivially, and together with it has the full correlation-group order. This gives the asserted direct products. In every other row conjugation is faithful on the full correlation group and identifies it with G or A.

Attribution and verification boundary. The classification, source coordinates, shadow criterion and normalizer theorem are reused from the reviewed material. The contribution here is the fifteen explicit applications, the evaluation of pair counts using transitivity of a single shadow, and the use of a perfect subgroup with abelian quotient to verify the required characteristic subgroup. These are elementary deductions; no claim of historical novelty is made. GAP's exact group algorithms and nauty's complete upper bounds remain computational premises. Literal small coset models check (1) against direct incidence counts, including repeated shadows. Source pages 3,4,8,9,11,18,19,20,21 were inspected in the retained edition. Abstract parabolic and Borel names, other geometries and the whole manuscript remain outside this component. No conjecture resolution, new Lean verification or compression measurement is asserted.

Coset coordinates and the last six Higman–Sims geometry groups

Let $G = \mathrm{HS}$, with $|G| = 44\,352\,000$, and let $A = \mathrm{Aut}(G)$, with $|A| = 88\,704\,000$. We acquire the reviewed faithful models on 100 points, including $G' = A' = G$ and $C_A(G) = 1$. The fixed source is the 23-page supplement dated 6 October 2003 accompanying D. Leemans, *The residually weakly primitive geometries of HS*, Australasian Journal of Combinatorics 33 (2005), 231–236. Its classification and the exact correspondence between source diagrams and saved parabolic families are premises of this application.

Theorem 1. *For the six specified families, the full groups are as follows. Here n is the smallest type size, a lists every model type of that size, d is the valency of the intrinsic graph, and q is the order of its full automorphism group divided by $|G|$.*

Family	*Source*	a	n	d	q	$\mathrm{Aut}(\Gamma)$	$\mathrm{Cor}(\Gamma)$
1	*2.6*	*2*	*36960*	*24*	*2*	A	A
2	*2.5*	*1*	*11550*	*64*	*2*	A	A
10	*3.27*	*2,3*	*15400*	*180*	*1*	G	A
16	*3.21*	*2,3*	*15400*	*24*	*2*	A	$A \times C_2$
17	*3.20*	*1,3*	*15400*	*24*	*2*	A	$A \times C_2$
18	*3.19*	*1*	*15400*	*24*	*2*	A	A

In the retained source, 3.20 prints $\mathrm{Aut}(\Gamma) = G$ *and* $\mathrm{Cor}(\Gamma) = A$*, while 3.21 prints* $\mathrm{Aut}(\Gamma) = G$ *and* $\mathrm{Cor}(\Gamma) = G \times C_2$*. The theorem replaces both entries in each of these two diagrams. Pages 7 and 8 were inspected directly. The four other rows agree with the printed groups, using the source convention when a correlation group is omitted. Previously recorded corrections to a type size in 2.6 and an incidence number in 2.5 are separate.*

Proving the point action from literal cosets. Fix a family $P_1, \dots, P_r$ and a smallest type a. Put $P = P_a$ and $n = [G : P]$. Let $g_1, \dots, g_t$ be the ordered generators of the original 100-point model. The certificate supplies elements $c_1, \dots, c_n \in G$ and permutations $r_1, \dots, r_t$ of $\{1, \dots, n\}$. The reader verifies

$$c_i g_j c_{i^{r_j}}^{-1} \in P \qquad (1 \le i \le n,\ 1 \le j \le t). \tag{1}$$

It also verifies transitivity of $\langle r_1, \dots, r_t \rangle$, the existence of a label b with $c_b \in P$, and $\mathrm{Core}_G(P) = 1$ in the original model.

The map $i \mapsto Pc_i$ respects each generator by (1). Every coset is reached from P by a positive generator word, since G is finite. Thus this map is surjective; its domain and codomain both have n elements, so it is bijective. Under this bijection the r_j are exactly right multiplication by g_j. Consequently they define a homomorphism $\phi : G \to E = \langle r_1, \dots, r_t \rangle$, whose kernel is $\mathrm{Core}_G(P) = 1$. The image is E by definition. Only after these checks does the program install the established homomorphism and its injectivity, surjectivity and order attributes in GAP. There is no unchecked homomorphism premise. Across the nine anchors, all 2 813 580 generator edges are checked.

Recovering objects from points. Write $E_b = \phi(P_a)$. A type-j base object has point shadow $S_j = b^{\phi(P_j)}$. Direct shadows suffice except at the two anchors of family 10, where type 1 requires additional incident objects. An encoding consists of its shadow together with typed sets of previously encoded incident objects. The reconstruction order is acyclic. For a reference to type k, the reader constructs the entire orbit of its base encoding under $\phi(P_j)$ and checks that its size is $[P_j : P_j \cap P_k]$. Induction then identifies that orbit with all incident type-k objects, with none omitted or duplicated.

Lemma 2. *Let a group E act on a set of objects equipped with an equivariant carrier in a point set. Suppose b belongs to the carrier S of an object e, and $K \leq E$ fixes e and is transitive on S. Then*

$$E_e = \langle K, (E_b)_e \rangle.$$

Proof. If $h \in E_e$, then $b^h \in S$. Choose $k \in K$ with $b^k = b^h$. The element hk^{-1} fixes both b and e, so $h \in (E_b)_e K$. The reverse inclusion follows because both factors fix e. □

For each recorded encoding the program checks its exact shadow, every referenced orbit, invariance under all generators of $\phi(P_j)$, and membership in $\phi(P_j)$ of every generator of its stabilizer inside E_b. Lemma 2 therefore gives full stabilizer $\phi(P_j)$. The orbit of the encoding has size $[G : P_j]$, so the map from type-j objects to encodings is injective. At family 10, anchor 2 reconstructs type 1 using types 2 and 3, and anchor 3 reconstructs type 1 using type 2. All remaining encodings use their direct shadows. The verifier requires every type to be reached. A typed automorphism fixing all anchor points then fixes every encoding, by induction, and hence every object. Its restriction to the anchor is faithful.

The intrinsic relation. We acquire the previously proved common-incidence formula. If O is a non-diagonal orbit of E_b on points, the number of type-j objects incident with a specified pair in the corresponding orbital is

$$m_j(O) = \frac{[G : P_j]|S_j|\,|S_j \cap O|}{n|O|}.$$

This formula counts objects, including distinct objects with equal shadows. The proof counts triples of an object and an ordered pair in its shadow; transitivity of P_j on that shadow makes the number of second points constant. For each pair take the sorted multiset of $([G : P_j], m_j(O))$ over all types. Correlations preserve it, including multiplicities. Order the distinct off-diagonal signatures lexicographically and choose the first whose relation has the indicated valency d. Every chosen relation is checked to be symmetric and irreflexive. The selection rule is intrinsic, so a correlation taking one anchor to another takes its graph to the other anchor graph. All smallest types are included.

The base neighbour set is propagated along the verified point generators. The program visits all n points and checks every adjacency row under every generator afterwards. Thus possible conflicting propagation paths cannot silently produce another graph. All rows have valency d.

Complete bounds and an explicit outer action. Let D be the full automorphism group of one of these graphs. A complete unseeded Traces calculation supplies $|D| = q|G|$ as in the table. The graph reader checks the whole adjacency array and all known E generators before calling Traces; it checks every returned generator afterwards. Only exact integer orders below 2^{53}, with zero exponent and clean completion status, are accepted. This is a computational upper bound, not a deduction from the returned generators alone.

When $q = 1$, $E \le D$ and equal orders imply $D = E$. When $q = 2$, we identify the extension directly. In the original model put $B = \bigcap_j P_j$ and $S = \bigcap_j N_A(P_j)$. The calculation verifies $S \cap G = B$, $|S| = 2|B|$, and an element $x \in S \setminus G$ with $x^2 \in P_a$. The latter element exists in each of the seven instances where $q = 2$. Starting with $b^u = b$, it propagates a point permutation u using the equations

$$(i^{\phi(g_j)})^u = (i^u)^{\phi(g_j^x)}.$$

Every encountered equation is checked, every point is reached, and the result is required to be bijective. The verifier then checks, as literal permutation identities,

$$\phi(g_j)^u = \phi(g_j^x) \quad \text{for every } j, \qquad u^2 = \phi(x^2).$$

It also checks u on every adjacency row of the actual graph.

These identities show that u normalizes E and $u^2 \in E$. Moreover, $u \notin E$. Otherwise $u = \phi(g)$ would give the same automorphism of G as x, so $xg^{-1} \in C_A(G) = 1$, contrary to $x \notin G$. Consequently $F = \langle E, u \rangle$ has order $2|G|$. Since $F \le D$ and the complete upper bound gives that same order, $D = F$. In both cases $E \lhd D$ and D/E is abelian. Perfection of G yields $E = E' \le D' \le E$, hence $D' = E$.

From graph groups to geometry groups. The preceding equality identifies the specified G intrinsically at every possible image anchor. If c is a correlation, conjugation by its point restriction carries one graph's derived group to the other. For every $g \in G$, the restriction of cgc^{-1} to the image anchor therefore equals that of some $h \in G$. The typed automorphism $h^{-1}cgc^{-1}$ fixes that anchor pointwise and is trivial by the faithful restriction just proved. Thus G is normal in the full correlation group.

The acquired coset normalizer theorem is now applicable. Let T be the setwise stabilizer of the parabolic family in A, allowing all type permutations. It gives

$$|\operatorname{Aut}(\Gamma)| = [G : B]|S|, \qquad |\operatorname{Cor}(\Gamma)| = [G : B]|T|.$$

For each type permutation, successive exact transporter intersections retain precisely the automorphisms carrying all already assigned parabolics to their targets. Positive representatives are checked on the entire family. Necessary subgroup-order tests discard only impossible permutations; no successful representative is inferred from an order.

Family	$\lvert B\rvert$	$\lvert S\rvert$	$\lvert T\rvert$	$\lvert T \cap G\rvert/\lvert B\rvert$	$\lvert \mathrm{Cor}(\Gamma)\rvert$
1	200	400	400	1	88704000
2	240	480	480	1	88704000
10	16	16	32	1	88704000
16	32	64	128	2	177408000
17	32	64	128	2	177408000
18	40	80	80	1	88704000

In each row $S \cap G = B$. The typed group therefore acts faithfully by conjugation on G and is identified with G or A by its order and its actual embedding in $\mathrm{Aut}(G)$. The full centralizer is the acquired quotient $(T \cap G)/B$. In families 16 and 17 it has order two, is central, and meets the typed group trivially. Their product has the full correlation group order, proving $A \times C_2$. In the other four rows that centralizer is trivial; conjugation faithfully embeds the full group in A and its order identifies it with A. These are identifications by explicit normalizer and centralizer arguments, not order-based group recognition.

Controls and failed computations. Literal subgroup enumeration in S_3, D_8, A_4 checks all coset edges, the carrier stabilizer lemma, and the incidence formula against direct counts, including repeated shadows. A separate example shows that carrier transitivity cannot be omitted from Lemma 2. Actual damaged certificates change a coset representative, a point generator, a shadow, and an acyclic type reference; each must be rejected. Graph controls change domains, edges and permutations. Ordinary and optimized Python must agree on all mathematical controls.

The Traces interface is additionally checked on every graph on four labelled vertices against all 24 permutations, including equality with the group generated by its output. Nonempty input generator lists are forbidden. An earlier interface that supplied known generators failed this control: for the complete graph on four vertices with one supplied transposition it reported order 18 instead of 24. That program, its input and the diagnostic are retained and supply no upper-bound premise here. All nine accepted bounds are fresh unseeded computations. Earlier capped nauty attempts, an overrestrictive preliminary outer-normalizer test and an invalid time-cap request are also retained without mathematical credit.

Attribution and remaining scope. The original geometry classification and source assertions are attributed to the retained manuscript and supplement. The coordinate correspondence, shadow reconstruction framework, common-incidence count and coset normalizer theorem are acquired from separately reviewed work. This component contributes the literal coset-edge proof, the use of a point stabilizer to test an object's full stabilizer, the explicit outer action, and their nine concrete applications. These are elementary arguments; historical novelty is not claimed. GAP's exact group algorithms and Traces's complete upper bounds remain computational premises. No Lean verification of this component is claimed.

The six full-group determinations complete this particular part of the 86-family analysis, conditional on the acquired classification. Abstract parabolic structures, the other uncompleted source clauses, and final whole-manuscript review remain separate. There is no new conjecture resolution or compression measurement in this component.

Two explicit Higman–Sims incidence constructions

This component verifies the two constructions in Section 4 of Leemans [1]. It uses the literal Higman–Sims group and the accepted finite coset catalogue, and provides explicit maps between that catalogue and the set constructions. The historical Higman model and its identification with HS are attributed in [1] to Higman and Sims. We do not replace that attribution by a claim of a new construction.

Our retained group input is a faithful permutation model $G \cong \mathrm{HS}$ of order 44 352 000, together with its complete automorphism action of order 88 704 000. The two previously recognized coset families have parabolic orders

$$(2880, 252000, 40320), \qquad (2880, 252000, 252000). \tag{1}$$

Their catalogue identifiers are 35 and 36. These identifiers refer to the acquired catalogue, not to numbering in the historical supplement. We retain the classical identification of the unique Moore graph of valency seven on fifty vertices as the Hoffman–Singleton graph, and its automorphism group $U_3(5) : 2$, as used in [1, Section 4]. The exact finite permutation and graph algorithms used in the certificate are stated with their software inputs; group orders alone are never used to identify an unnamed abstract group.

Theorem 1. *There are sets $\mathcal{P}, \mathcal{C}, \mathcal{Q}$, of respective sizes $176, 1100, 176$, in the retained literal model, with conics $C \in \mathcal{C}$ eight-subsets and quadrics $Q \in \mathcal{Q}$ fifty-subsets of $\mathcal{P}$, satisfying all six Higman properties printed in [1, Section 4.1]. Put $\mathcal{L} = \binom{\mathcal{P}}{2}$.*

1. *The geometry Γ_C on $\mathcal{P}, \mathcal{L}, \mathcal{C}$, with incidence given by containment, is isomorphic to family 35. Its point residue is the Hoffman–Singleton graph, its conic residue is the vertex–edge geometry of K_8, and its line residue is a digon. It has $\mathrm{Aut}(\Gamma_C) = \mathrm{Cor}(\Gamma_C) = G$.*

2. *In Γ_Q on $\mathcal{P}, \mathcal{L}, \mathcal{Q}$, point incidence is containment. A line is incident with a quadric precisely when the quadric contains both conics through its two points. This geometry is isomorphic to family 36. Both point and quadric residues are Hoffman–Singleton graphs, and its line residue is a digon. Its type-preserving group is G, and $\mathrm{Cor}(\Gamma_Q) \cong \mathrm{Aut}(\mathrm{HS})$.*

Both geometries have 61 600 chambers and Borel subgroup S_6. Their line-residue kernels are S_6, and their line stabilizers are split extensions $S_6 : 2^2$. Their point stabilizers, and the quadric stabilizers of Γ_Q, act faithfully as the full $U_3(5) : 2$ on their Hoffman–Singleton residues. The conic stabilizer of Γ_C acts faithfully as S_8 on its eight points.

Literal Higman data and the sixth property

Take the right-coset action of G on a specified subgroup of order 252000. The saved generator images give a homomorphism with trivial kernel on the resulting 176 labelled points. A saved element of the complete ambient automorphism action conjugates the second family's point parabolic to the first family's point parabolic. This actual element, all three transported subgroups, and all point-generator images are checked. Thus the two coordinate systems agree by an explicit map.

The conics and quadrics are the full orbits of the specified base eight-set and fifty-set. Every generator preserves the complete lists and their incidence relations. Each of the four sets of objects (points, lines, conics and quadrics) is a single orbit. Literal subset calculations give fifty quadrics through each point and eight quadrics through each conic. For each of all 15400 point pairs there are exactly two conics C_1, C_2, and

$$\{Q : p, p' \in Q\} = \{Q : C_1 \subseteq Q\} \cup \{Q : C_2 \subseteq Q\}, \qquad |\{Q : C_1 \cup C_2 \subseteq Q\}| = 2. \tag{2}$$

The two quadrics in the last set give a bijection from point pairs to quadric pairs. Conversely, every quadric pair contains exactly two conics; its point intersection is their union, and their intersection has size two. These checks establish properties (i)–(v), including both universal alternatives in (iv) and (v).

For property (vi), fix the retained conic S. For each of its eight points p and each of its eight incident quadrics Q, test both conditions in the source: every conic meeting S in two points including p is contained in Q, and every conic contained in two quadrics through S, one of which is Q, contains p. The complete 8 by 8 truth table has exactly one simultaneous solution in each row and column. With the retained labels the matching is

$$(1,6),(26,37),(36,23),(61,34),(165,9),(167,17),(170,30),(174,41).$$

The first coordinate is a point label and the second a quadric index. Since the conditions use only incidence and G is transitive on conics, this proves (vi) for every conic. The actual actions on points and on quadrics are doubly transitive; the point stabilizer has quadric orbits of lengths 50 and 126. This also proves the source's transitivity assertions on incident and nonincident point–quadric pairs.

Coset maps, flags, and residues

For either family write H_1, H_2, H_3 in line, point, plane order. The retained base objects b_i have exactly these stabilizers in G. For every coset the certificate records the map

$$H_i g \longmapsto b_i^g. \tag{3}$$

It verifies each image literally and verifies the resulting six full bijections. Stabilizer equality makes (3) well-defined and equivariant. Two right cosets are incident when they intersect. The neighbours of H_i of type j are therefore the orbit of H_j under H_i. For all twelve ordered pairs of distinct types in the two families, the actual orbit is compared to the entire geometric neighbour set of b_i. Equivariance transports these equalities to all objects. This proves both incidence isomorphisms, without inferring incidence from matching degrees alone.

Every line has two points and two incident planes, and all four point–plane flags in its residue occur. Every incident point–plane pair extends through exactly seven lines. Thus every partial flag extends to a chamber, and both geometries have $15400 \cdot 4 = 61600$ chambers. Their base chamber stabilizers have order 720 and have explicit bijective generator maps onto Sym(6). Their G-orbits therefore contain all chambers. This is actual flag transitivity. The previously recognized residual connectivity, local primitivity, weak primitivity and intersection property transfer across (3).

For Γ_C, the base point residue has fifty conics as graph vertices and 175 lines as graph edges. All edges have distinct two-element endpoint sets. Breadth-first calculations at every vertex give valency seven, diameter two and girth five. The same calculation holds for the point and quadric residues of Γ_Q. The retained uniqueness statement identifies these graphs as Hoffman–Singleton graphs. The conic residue of Γ_C is literally its eight points and all their 28 pairs, hence K_8. Line residues are the complete bipartite graph on two points and two planes. Object transitivity transports the base-residue conclusions to all residues.

The faithful actions of the three order-252000 parabolics on their fifty-vertex residues have image order 252000. They therefore equal the full known automorphism group of the Hoffman–Singleton graph. The conic stabilizer's action equals Sym(8) and has trivial kernel. For each line stabilizer, the kernel on the four residue objects is exactly the already identified Borel S_6. Its image is elementary abelian of order four. The certificate supplies two commuting involutions generating an order-four subgroup disjoint from this kernel; together they generate the full line stabilizer. This proves the splitting asserted by the colon notation, in addition to the kernel and quotient orders.

The full automorphism and correlation groups

For Γ_C, restrict an automorphism to the bipartite point–conic incidence graph, with its two vertex classes distinguished. This restriction is faithful: a line is determined by its two points. Every graph

automorphism extends uniquely to lines, and the certificate checks all returned generators on every line–conic incidence. The complete graph calculation, on 1276 vertices, gives order 44 352 000. The explicit G-generators lie in that group and generate a faithful image of the same order, so equality holds. Since the three geometry types have distinct cardinalities, every correlation preserves types. Hence $\mathrm{Cor}(\Gamma_C) = \mathrm{Aut}(\Gamma_C) = G$.

For Γ_Q, use the point–quadric incidence graph, adjoining one marker for each type. The two markers form one separate colour class; each is joined to all 176 objects of its type. The 354-vertex graph therefore allows precisely type-preserving or type-interchanging permutations, not arbitrary permutations of incidence types. The complete group has order 88 704 000. Every returned generator is checked on all graph adjacencies. It has a unique lift to lines: if types are fixed, use the image point pair; if types are interchanged, use the inverse of the point-pair/quadric-pair bijection in (2). In both cases all 15400 resulting line incidences are checked, so these lifts are correlations of the full geometry. Conversely every correlation restricts to this graph, and the restriction is faithful.

Let D be the resulting group and let E be the explicit image of G. The exact generator map $G \to E$ is bijective. The kernel of D on the two markers equals E, its marker image has order two, and $C_D(E) = 1$. Thus conjugation embeds D in $\mathrm{Aut}(E)$; its order equals the retained full order of Aut(HS), so this embedding is surjective. Consequently the type-preserving group is G and the correlation group is Aut(HS). A saved generator interchanges the types and provides the duality claimed in [1, Section 4.2].

Certificate boundary

The finite files retain the complete object lists, both generator actions, six coset maps, twelve neighbour comparisons, the sixth-axiom matching, all small-graph generators and their line lifts, and explicit Borel and complement maps. Completeness of the graph-automorphism calculations is an acquired nauty premise; the returned generators are additionally checked literally. The GAP finite group and homomorphism algorithms and the retained identification of the ambient HS model are explicit dependencies. The original failed test of surjectivity onto the declared range of an action homomorphism is preserved. The successor uses its trivial kernel, the correct faithfulness assertion.

This proves the two construction and group descriptions in Section 4 of the fixed six-page article. It does not assert completion of the article's enumeration, Table 1, or historical supplement labels. Those obligations belong to their separately scoped components.

References

[1] D. Leemans, *The residually weakly primitive geometries of HS*, fixed six-page source, pp. 231–236, Section 4, pp. 233–235.

[2] G. Higman, *On the simple group of D. G. Higman and G. C. Sims*, Illinois J. Math. **13** (1969), 74–80. The historical attribution is retained as stated in [1].

Five explicit Higman–Sims doublings and three source-link qualifications

L091 finite construction component

We use the archived supplement dated 6 October 2003 to D. Leemans [1], with its accepted coordinates in the 86-class literal coset catalogue. All labels below are supplement labels. The main paper calls the construction of [2, Theorem 4.1] doubling. We use precisely the two-point-shadow operation in the acquired theorem adapter: the prior primary reading included that theorem and its construction, but did not obtain the complete article. No different convention is silently imposed. The elementary argument needed here is given in full.

Theorem 1. *The following five printed links have explicit incidence isomorphisms from the indicated doubled parents to the target geometries:*

$$3.2 \longrightarrow 3.3, \quad 3.4 \longrightarrow 3.5, \quad 3.8 \longrightarrow 3.9, \quad 4.8 \longrightarrow 4.10, \quad 4.14 \longrightarrow 4.19.$$

The other three printed links, $3.12 \to 3.15$*,* $3.13 \to 3.16$ *and* $3.20 \to 3.21$*, are not applications of this operation to the stated parents. The first two fail its two-point hypothesis at the only type choice compatible with the target sizes; the third fails the necessary type-size condition for every choice.*

These are ancillary construction qualifications, relative to the explicit operation below. They neither refute the principal classification nor claim that the three target geometries do not exist. Their existence and coordinates remain acquired. No repair using an unspecified alternative doubling convention or an unverified replacement parent is asserted.

1 The construction and exact finite test

Lemma 1. *Suppose a flag-transitive geometry has types* X, E *such that the* X*-shadows of its* E*-objects are distinct two-element sets. Suppose, for any other object* z*,*

$$e * z \quad \Longleftrightarrow \quad \sigma_X(e) \subseteq \sigma_X(z). \tag{1}$$

Replace X, E *by two copies* X^+, X^- *of* X*. Join* x^+ *to* y^- *exactly when* $\{x, y\}$ *is an* E*-shadow; join either copy of* x *to* z *exactly when* $x * z$ *originally. Leave other incidences unchanged. This is a geometry with the same chamber action as the original one.*

Proof. An old chamber (x, e, F) determines the new chamber (x^+, y^-, F), where y is the other endpoint of e. Conversely a new flag using both copies determines its unique edge, and (1) makes that edge incident with all its unchanged objects. It therefore extends from an old flag to an old chamber and then to a new chamber. A flag using only one copy extends from the corresponding old point flag; if neither copy occurs, extend its unchanged flag first. This proves chamber extension in every case and the asserted equivariant chamber bijection. Chamber transitivity then implies flag transitivity. The argument also works in rank two; it does not assert extra rank-two connectivity hypotheses from a theorem stated only in higher rank. □

Write $G = \mathrm{HS}$. Objects are right cosets; incidence means nonempty intersection. We also denote the base point and edge stabilizers by X, E. The number of points incident with E is

$[E : E \cap X]$. When this is two, choose a recorded $h \in E \setminus X$; the endpoints are X and Xh. Their pointwise stabilizer is $X \cap X^h$, where $X^h = h^{-1}Xh$. Hence distinct edge shadows are equivalent to

$$2|X \cap X^h| = |E|. \tag{2}$$

Indeed, E is contained in the pair's setwise stabilizer and interchanges its endpoints. That setwise stabilizer has twice the order of the pointwise stabilizer, so (2) gives equality. The resulting map from E-cosets to pairs is therefore injective.

For any remaining base stabilizer Z, take right transversals T_X of $X \cap Z$ in X and T_E of $E \cap Z$ in E. The following finite conditions are equivalent to (1):

$$ha^{-1} \in XZ \iff a^{-1} \in EZ \qquad (a \in T_X), \tag{3}$$

$$a^{-1} \in XZ \text{ and } ha^{-1} \in XZ \qquad (a \in T_E). \tag{4}$$

Here XZ and EZ are identity double cosets, not asserted subgroups. The objects Za, $a \in T_X$, are exactly the Z-objects incident with X. The second endpoint meets Za precisely when $ha^{-1} \in XZ$, while E meets it precisely when $a^{-1} \in EZ$. This proves sufficiency of a pair of endpoints in (3). Equation (4) checks necessity for every Z-neighbor of E, including those not assumed to meet X. Transitivity transports these tests to all edges. No enumeration of the 44 352 000 ambient elements is needed.

2 Five literal target maps

The new geometry has base stabilizers

$$X,\ X^h,\ (Z_i)_{i \neq X,E}. \tag{5}$$

In the minus copy the base point is Xh, so the right coset $X^h g$ maps to Xhg. The chamber argument makes every type-pair incidence a single G-orbit, and thus (5) is its full coset description. Each certificate supplies an actual n in the acquired normalizer N of G and a type permutation, and checks that conjugating every subgroup in (5) by n gives the target family in that order. Consequently $Hg \mapsto H^n g^n$ is a well-defined typewise bijection preserving incidence and nonincidence. This is an exact incidence map, not an inference from a diagram or subgroup order.

The following table uses internal catalogue type indices solely to make the literal certificates easy to locate. The source-coordinate ledger also supplies the corresponding native node indices.

Parent→target	Internal parent→target	(X, E)	(3) tests	(4) tests
$3.2 \to 3.3$	$35 \to 34$	$(3, 1)$	8	2
$3.4 \to 3.5$	$32 \to 33$	$(3, 2)$	15	3
$3.8 \to 3.9$	$30 \to 28$	$(3, 1)$	8	2
$4.8 \to 4.10$	$70 \to 69$	$(4, 3)$	2310	48
$4.14 \to 4.19$	$54 \to 57$	$(2, 3)$	2800	55

The two rank-four rows test both unchanged types. The accepted target geometries transport all their residue properties through these maps. The present component does not infer full automorphism or correlation groups merely from the induced specified G-action.

3 Three exact obstructions and evidence scope

For the links $3.12 \to 3.15$ and $3.13 \to 3.16$, the parent and target size lists, in the same order, are

$$(352, 30800, 1100) \longrightarrow (352, 1100, 1100),$$
$$(352, 30800, 5600) \longrightarrow (352, 5600, 5600).$$

Thus the only possible choices duplicate the 1100- or 5600-object type and remove the 30800-object type. In each case the edge stabilizer has order 1440, and its intersection with the proposed point stabilizer has order 144. Each edge shadow therefore has ten points, not two. These are literal subgroup intersections, also acquired in the numerical metadata. In particular a type permutation cannot fix the obstruction.

For $3.20 \to 3.21$, the parent and target size lists are

$$(15400, 346500, 15400) \quad \text{and} \quad (30800, 15400, 15400).$$

The operation replaces two sizes by two copies of one existing size and retains the third. It cannot create the new size 30800. The native statements are on supplement pages6,7 and8; the eight links and their source-node maps are explicitly retained.

The checker reads the exact previously accepted group, complete family and source-coordinate inputs. GAP's finite subgroup, transversal and permutation computations remain explicit computational premises. It verifies all five saved subgroup maps directly, all local tests (2)–(4), and all three obstructions. Thirty damaged certificates are rejected. An independent literal-coset control over S_3, D_8, A_4 compares the criterion with all actual pair shadows and all relevant edge–object incidences, including both successful and failed endpoint conditions. The failed first probe is preserved: it returned three positive cases before stopping at the first ten-point shadow. The changed probe records all eight outcomes; default replay performs certificate checks, not a new classification or transporter search. Source transcription, prior classifications and their retained subgroup-data premises are acquired without additional census credit.

References

[1] D. Leemans, The residually weakly primitive geometries of HS, Australas. J. Combin. **33** (2005), 231–236, and the retained 23-page supplement dated 6 October 2003.

[2] C. Lefèvre-Percsy, N. Percsy and D. Leemans, New geometries for finite groups and polytopes, Bull. Belg. Math. Soc. Simon Stevin **7** (2000), 583–610.

Inner conjugacy and residual primitivity in the HS supplement

A finite source component using acquired subgroup data

We use the fixed six-page article by Dimitri Leemans, *The residually weakly primitive geometries of HS*, Australasian Journal of Combinatorics 33 (2005), 231–236, and its retained 23-page supplement dated 6 October 2003. Diagram numbers below refer to that supplement. This component checks its conjugacy notes, three stronger primitivity annotations, and parenthetic numbers. It does not assert whole-source completion.

The acquired model is $G = \mathrm{HS}$ on 100 points, of order 44352000. The accepted input contains 589 representatives of subgroup classes and 3075 actual maximal-inclusion pairs (K, M), with each parent K one of these representatives. Their completeness and the named ambient model are retained input premises, already acquired by the completed geometry classification. We use the entire maximal-inclusion list, including inclusions outside its separate locally two-transitive sublist. The accepted classification supplies 86 literal families $(P_1, \ldots, P_r)$; the independently accepted source-coordinate bijection supplies the ordered source-to-model type permutation. The accepted parabolic structure packet supplies the group names.

Proposition 1. *All 24 subgroup-pair relations specified by the 18 conjugacy notes listed below hold inside G, with the source's actual type coordinates.*

Proof. For each pair (P, Q) the certificate applies the complete subgroup conjugation algorithm `RepresentativeAction(G,P,Q,OnPoints)`. Every positive result is retained as an actual permutation $x \in G$, and the replay verifies $P^x = Q$. There are eleven positive results. For each of the thirteen negative results the exact complete algorithm returns failure; the replay checks that failure again. Completeness of this exact permutation-group algorithm is an explicit software premise. No claim is inferred merely from equality of group orders or names, nor is conjugation in $\mathrm{Aut}(G)$ substituted for conjugation in G.

The source join fixes both ordered type indices before this test. Thus even repeated abstract names are checked at their actual occurrences. For 4.41 the top pair and bottom pair are each conjugate, and all four cross pairs are nonconjugate. Diagram 5.6 has two separate nonconjugacy claims, for M_{11} and A_7 respectively. □

Name	Conjugate in G	Nonconjugate in G
M_{11}	3.16, 3.30, 4.15	2.4, 4.39, 4.40, 5.6, 5.7
$\mathrm{U}_3(5) : 2$	4.9, 4.10, 4.25, 4.26	3.1
$\mathrm{U}_3(5)$	4.16, 4.19	4.11
A_7	—	5.6, 5.8

The additional six M_{11} relations of 4.41 are as just specified. Their literal source nodes and transported model types are in `claims.json`; they are not reconstructed from a drawing layout or abstract group identifier during replay.

Proposition 2. *The displayed geometries 3.1, 3.2 and 4.1 are residually primitive for the specified G action: every nonempty type of every residue carries a primitive action of its flag stabilizer.*

Proof. Write $I = \{1, \dots, r\}$, $H_J = \bigcap_{j \in J} P_j$, with $H_\varnothing = G$. Flag transitivity, acquired for each literal geometry, transports every flag of type J to the base flag. For $i \notin J$, its incident objects of type i form the transitive H_J-set with point stabilizer $H_{J \cup \{i\}}$. A transitive coset action is primitive exactly when this stabilizer is a maximal proper subgroup. The statement does not require the action to be faithful: subgroups between a point stabilizer and the acting group correspond to blocks through its base point, and quotienting by the action kernel does not change that interval.

For each required inclusion $L = H_{J \cup \{i\}} < H = H_J$, the certificate retains an index a in the acquired 589-class roster and $x \in G$ with $H^x = K_a$. It also retains an actual maximal-inclusion row (K_a, M) and $y \in K_a$ with $(L^x)^y = M$. Both equalities, both element memberships, and proper containment are verified literally. Since y normalizes K_a, conjugation by xy identifies (H, L) with (K_a, M). The acquired maximality of M in K_a proves the result. The checker enumerates every proper $J \subset I$ and every $i \notin J$, including $J = \varnothing$ and rank-one residues. Hence there are $r2^{r-1}$ inclusions: twelve each for representatives 36 and 35 (diagrams 3.1 and 3.2), and thirty-two for representative 77 (4.1). All fifty-six have positive literal transports. This pays the all-type claim, which is stronger than the acquired weakly primitive condition. □

Proposition 3. *For each of the 86 displayed diagrams, its retained parenthetic number (one when absent) equals the number of acquired inner/type family classes in its fusion fiber under full automorphisms and type permutations. The individual numbers comprise 42 ones and 44 twos, summing to 130.*

Proof. The acquired complete fusion file contains 130 member records with distinct input identifiers and 86 representative identifiers. Its producer explicitly stores the representative as `source` and the original input as `target`. Count records by this source identifier, then apply the accepted source-coordinate bijection. `literal_join.py` requires every identifier, every diagram, and every individual printed number to agree. The fiber counts concern the specified action and its inner equivalence. They are not additional bare incidence classes; the accepted bare classification has 86. □

The finite derivation returned cleanly without an ambient subgroup or geometry census. Its completed rows, exact loaded inputs and terminal receipt are retained. Independent small controls exhaust all subsets of the elements of C_4, D_8, S_3 to form their subgroup sets, test maximality by all intervening subgroups, and test conjugacy by every acting element: 159 comparisons, including negative cases. The replay also rejects damaged source coordinates, fabricated maps, wrong maximal-inclusion indices and reversed fusion directions. An initial control harness emitted unbound-global warnings; its clean successor uses the exact helper prefix without the unused target entry point. A Python symlink-locator mismatch was rejected before the actual call and repaired without changing any mathematical input.

The three-RPRI, 24-conjugacy and 86-number statements are the complete scope of this component. The eight-group comparison table and final source synthesis remain separate, as do already qualified doubling and attribution clauses. No new census total, conjecture, measurement or whole-original admission is assigned here.

Exact joins for the eight sporadic rows in the Higman–Sims table

Source recovery component

The fixed source is D. Leemans, *The residually weakly primitive geometries of the Higman–Sims group*, Australasian J. Combin. 33 (2005), 231–236, Table 1 on p. 232. We join existing complete computations and explicit source coordinates. No subgroup or geometry census is repeated. The last row retains the qualifications below; this component does not claim completion of the whole article.

1 Subgroup counts from complete retained inputs

For a representative H_i of a G-conjugacy class, its class length is $\ell_i = [G : N_G(H_i)]$. Thus the total number of subgroups is $\sum_i \ell_i$, provided that the representative list is complete. The six atlas inputs for $M_{11}, M_{12}, J_1, M_{22}, J_2, M_{23}$ have separately accepted complete numerical tables, with literal class-length and source-row correspondences. Their exact review records, all rows and required dependencies are acquired. No abstract name is inferred from an order or from matching the source total.

Proposition 1. *In a complete table of marks, let $m_{i,1}$ be the mark at the trivial subgroup and $m_{i,i}$ the diagonal mark. Then*

$$\ell_i = \frac{m_{i,1}}{m_{i,i}}.$$

Proof. The transitive coset set associated with H_i has $[G : H_i]$ elements, all fixed by the identity. Its H_i-fixed cosets are precisely those represented by elements of $N_G(H_i)$: a containment between conjugates of the same finite order is equality. Hence the two marks are $[G : H_i]$ and $[N_G(H_i) : H_i]$. Their quotient is $[G : N_G(H_i)]$. Distinct conjugacy classes partition the subgroup set. □

For HS the retained complete 589-class input is already used by its accepted geometry census. A bounded extraction checked equality of all 589 subgroup orders and the literal ordered generating permutations against that exact table model. It returned every pair of marks, and compared their ratios with the library class lengths. The clean terminal execution is retained; the present Python check independently recomputes every ratio and the sum. Thus this is a new arithmetic join to an acquired complete input, not a new subgroup enumeration.

G	$\lvert G\rvert$	conjugacy classes	subgroups
M_{11}	7920	39	8651
M_{12}	95040	147	214871
J_1	175560	40	158485
M_{22}	443520	156	941627
J_2	604800	146	1104344
M_{23}	10200960	204	17318406
HS	44352000	589	149985646

These 21 fields agree with the fixed Table 1. In particular the printed J_1 total is 158485; an earlier project transcription of that number is not a source correction.

2 Geometry populations and truncations

The retained panel-recognition and maximal-deletion induction enumerate firm, residually connected, flag-transitive, RWPRI coset geometries with $(\mathrm{IP})_2$. The local $(2T)_1$ condition is imposed on the actual panel actions. For the phase-six M_{11}, M_{12}, J_1 calculations it is applied to the complete broader census afterward; for the phase-seven M_{22}, M_{23} calculations it is an input filter. Every Borel anchor has either no admissible panels or a terminal empty level. A finite rank cutoff without such evidence is not used as exhaustion.

Equivalence in these five retained computations is under $\mathrm{Aut}(G)$ and arbitrary type permutation. Their actual whole-family transporters and finite recognition tests are retained. The M_{11} certificate is expressly acquired here with its full completed execution and source artifacts; the prior L096 assembly's five-item census inventory did not itself list this certificate. We do not invent a separate earlier acceptance of it. The common recognized method, subgroup catalogues and exact finite-group software are explicit premises of the present reuse.

A truncation removes types from a higher-rank member of the *same* locally two-transitive class. This restriction matters: allowing higher members of the broader non-$(2T)_1$ class gives different answers, for example the M_{11} rank-four value becomes 5 instead of 11. Such a broader operation is not silently substituted for Table 1's class. Exact retained proper-subset tests and positive family maps determine the deleted classes. The checker joins the full witness counts, $(2T)_1$ flags, fusion partitions and retained truncation outputs; it acquires the underlying complete tests rather than rerunning them.

G	rank 2	rank 3	rank 4	rank 5	rank 6
M_{11}	0	9	11	0	0
M_{12}	5	10	17	1	0
J_1	11	1	2	0	0
M_{22}	1	20	12	9	0
J_2	5	16	3	0	0
M_{23}	0	3	16	10	0
HS	4	20	41	8	1

These are nontruncation counts in the acquired equivalence conventions. The accepted J_2 join uses 87 proper subsets, ten exact positive maps and six truncation targets. Its complete population is $8, 19, 3$ in ranks $2, 3, 4$. The HS result uses its separately accepted bare-incidence classification, all 756 proper subsets, 15 positive maps and twelve distinct truncation targets; its complete population is $6, 30, 41, 8, 1$. The HS bare-separation argument is not transferred to other groups without proof.

The source prints 11 for M_{12} rank three, whereas the exact retained specified-action calculation gives 10 under the class-restricted truncation convention. We record this precise comparison; no assertion that a different, unproved bare-incidence quotient has ten classes is needed. The missing M_{12} archive class identified in the prior accepted work fails local two-transitivity, so it cannot repair this entry. No nontruncation count is obtained merely by subtracting the number of witness maps: distinct target classes, not map multiplicities, are removed.

3 The remaining Janko boundary

The separately retained J3 supplement is dated 28 November 2003 and has thirteen pages. All its native pages were read for this source join. Its 137 printed subgroup rows sum to 71564248, which agrees with Table 1. This is exact arithmetic on a transcribed table, not a new proof that the

subgroup roster is complete or that every printed normalizer is correct. The cited classification is attributed to Leemans's J3 article; the subgroup table cites the earlier subgroup work. The main article's full native PDF was not acquired during this audit.

The supplement explicitly says "up to isomorphism". It prints seven rank-two diagrams, of which two are identified as truncations, and seventeen rank-three diagram labels. Four rank-three labels carry a multiplicity two, giving 21 supplied occurrences. The accepted literal archive analysis instead has 19 classes under $\mathrm{Aut}(J_3)$ and type permutation, with the precise source-label correspondence retained. Labels 3.7 and 3.8 each retain two such classes; the pairs at 3.11 and 3.13 are fused by actual outer maps. Equality of diagrams does not itself identify geometries.

Accordingly we do not import "exactly 17 classes" as a completeness premise consistent with that literal population. Nor does the existence of 19 specified-action classes automatically prove 19 bare-incidence classes. A proposed completeness premise for the full *occurrence* list, and the full-group normalization needed to settle that equivalence boundary, require an explicit separate disposition. The rank-two value 5, the absence of higher-rank J3 geometries and the 137-class subgroup completeness remain source-attributed here. No fresh J3 census is claimed.

4 Executable and attribution boundary

The check rehashes all acquired inputs and recomputes the 732 accepted atlas row lengths, 589 HS mark ratios, five certificate joins and the accepted J_2/HS target-class complements. Fourteen damaged joins must fail. Literal mark-diagonal controls for C_2, C_2^2 and S_3 check the arithmetic interface. There is no GAP census in this replay. The old finite programs, retained catalogue assumptions, explicit reviews and source images remain bound. The inherited induction and recognition theory is reused, not advertised as a new theorem. No original-source, conjecture, Lean or compression credit follows from this component alone.

The supplied third Janko geometries without a distinguished action

A finite application of the accepted incidence certificates

Let $G = J_3$, of order $N = 50\,232\,960$, and let Γ_i, $1 \le i \le 21$, be the literal rank-three coset geometries in the acquired archive. Indices refer to its final assignment, which exchanges construction entries 8 and 9. The accepted eligibility certificates prove that all twenty-one are firm, residually connected, flag-transitive, residually weakly primitive, locally doubly transitive and satisfy IP_2. An isomorphism below may permute the types and need not preserve the specified image of G.

Proposition 1. *The supplied geometries have nineteen isomorphism classes. The only identifications are* $\Gamma_{13} \cong \Gamma_{14}$ *and* $\Gamma_{16} \cong \Gamma_{17}$.

Proof. Write $G_J = \bigcap_{j \in J} G_j$ for a base flag stabilizer, with $G_\varnothing = G$. Flag transitivity implies that the number of flags of type J is $N/|G_J|$: the orbit map sends a right coset $G_J g$ to the translate of the base flag, and equality of translates is precisely equality of these cosets. Therefore the multiset of all eight numbers $N/|G_J|$ is intrinsic to the incidence geometry. A type permutation merely permutes the eight subsets. Since N is the same for every input, the corresponding multiset of eight stabilizer orders is also an intrinsic separating invariant. This argument does not assume that an arbitrary isomorphism normalizes G.

The acquired intersection orders give sixteen distinct profiles. Their only nonsingleton fibres are

$$\{7, 8\}, \qquad \{9, 10, 11\}, \qquad \{13, 14\}, \qquad \{16, 17\}.$$

The saved whole-family automorphisms, conjugators and type permutations give actual incidence isomorphisms for the last two pairs: a map carrying the subgroup family to its target induces a well-defined bijection of cosets and preserves nonempty intersections in both directions.

The accepted common point-graph argument separates 7 from 8 and 9 from 11 even without a distinguished group action. It proves that any such geometry isomorphism must normalize the point action of G: the point graph has full automorphism group of order $2N$, whose derived subgroup is the distinguished G, and the other objects have distinct point shadows. The incompatible involution counts in the resulting parabolics then exclude both isomorphisms. We use that complete argument, not merely its invariant under group automorphisms.

Finally Γ_{10} satisfies IP, whereas Γ_9 and Γ_{11} fail WIP. Both properties are expressed solely through flags, incidences and shadows, so are preserved by arbitrary isomorphisms and by type permutations. The implication IP $\Rightarrow$ WIP separates 10 from both other members of its fibre. Thus the first repeated fibre splits into two classes and the second into three, while each of the last two is one class. The number is $16 + 1 + 2 = 19$. □

Source use and the scope of the count

The seventeen diagram labels of the acquired Janko supplement have twenty-one printed occurrences. Labels 3.7 and 3.8 each account for two distinct classes; labels 3.11 and 3.13 each account

for one class despite their two supplied occurrences. The other thirteen labels each account for one. In particular, counting the seventeen labels is not counting the nineteen supplied incidence classes.

L091 is D. Leemans, *The residually weakly primitive geometries of HS*, Australasian Journal of Combinatorics **33** (2005), 231–236. Its introduction expressly treats the Janko classification as previous work, citing reference [9]: D. Leemans, *The residually weakly primitive geometries of* J_3, Experiment. Math. **13** (2004), 429–433 (our historical identifier L097). Its Table 1 in Section 3 quotes comparison counts from those earlier classifications. The new HS classification and the two Hoffman–Singleton constructions are separate claims of L091, supported by their own accepted proofs.

The proposition proves the exact quotient of the supplied Janko roster and, unconditionally on any global census, the existence of at least nineteen eligible rank-three geometries. It does not enumerate all Janko subgroup families. To conclude that nineteen is the global rank-three total, one must retain the earlier classification's coverage premise, in the explicit form that every eligible geometry occurs in this roster. To interpret all nineteen as nontruncations, one must also retain its absence-of-higher-rank premise. These are borrowed classification inputs, not conclusions of the proposition; the old numerical value seventeen is not imported as a premise. In particular no claim of an independently reproved global Janko census is made.

The full text of L097 is unavailable in the fixed accessible catalogue, and L097 is outside the active 105 mathematical originals. The acquired thirteen-page author supplement supplies auxiliary data and does not change that status. L096, whose supplied group families and intersection certificates are reused here, is an active original with an accepted corrected principal recovery. Neither use awards a second original or recovery credit to L097. The Janko supplement's other diagram metrics, abstract subgroup names and full automorphism groups are not new assertions of L091 and need not be reclassified to audit its comparison table. Existing positive certificates for some of them remain available.

The read-only verifier acquires the accepted finite inputs, reconstructs every flag-count profile and this quotient, and records the historical source correspondence separately. It performs no new group search. The accepted GAP and Traces calculations retain their stated software premises. This packet makes no Lean, conjecture or compression claim.

The third Janko row in the Higman–Sims comparison table

A source join of the acquired finite incidence certificates

Put $G = J_3$ and $N = 50\,232\,960$. The source of the comparison is Table 1, pages 232–233, of D. Leemans, *The residually weakly primitive geometries of HS*, Australas. J. Combin. **33** (2005), 231–236 (L091). Its Janko classification is expressly attributed to reference [9], the earlier Janko paper (L097). We distinguish the finite results below from that earlier classification's coverage premise.

Proposition 1. *The seven acquired rank-two prescriptions are eligible and mutually nonisomorphic, even allowing a permutation of the types. Exactly two are truncations of a supplied rank-three geometry:* 2.3 *is the truncation of active row* 2 *on types* $1, 3$*, and* 2.6 *is the truncation of active row* 1 *on those types. The other five are not truncations of any of the twenty-one supplied rank-three geometries.*

Proof. Eligibility of $2.1, 2.2, 2.4, 2.5, 2.7$ is already established by the accepted literal realizations. For 2.3 and 2.6 we use the actual subgroups P_1, P_3 of their accepted parent. The incidence maps are the identity on the retained right-coset sets, so preserve both incidence and nonincidence. Their intersections and indices are

	$\lvert P_1\rvert$	$\lvert P_3\rvert$	$\lvert P_1 \cap P_3\rvert$	$[P_1 : P_1 \cap P_3]$	$[P_3 : P_1 \cap P_3]$
2.3	8160	2880	480	17	6
2.6	1080	2880	180	6	16

The complete actual local coset actions are doubly transitive, hence primitive, and the two subgroups generate the accepted ambient group. Thus these geometries are connected and firm; all rank-one residues are nonempty. Rank-two coset incidence is flag-transitive because an incident pair has a common representative. For weak primitivity at the empty flag, retain the parent's already checked primitive type: type 1 in active row 2, and type 3 in active row 1. These are the same actual subgroups and ambient actions, not merely subgroups of the same order. Together with the primitive local actions this proves residual weak primitivity.

For completeness, the rank-two intersection test uses all double cosets. Put $B = P_1 \cap P_3$. The neighbours of P_1 are the cosets P_3h, where h runs over a right transversal of B in P_1. Two distinct objects of the first type with a common object of the second type can be transported to P_1, P_1x with $x \in P_3 \setminus B$. The orbits of such pairs are indexed by the nonidentity (B, B) double cosets in P_3. For every representative the two complete neighbour sets intersect in exactly one object, as checked in the actual finite records. Hence both geometries satisfy IP_2. No implication from local double transitivity to this intersection assertion is assumed.

The seven unordered pairs of object cardinalities are

2.1	17442	43605	2.2	17442	17442
2.3	6156	17442	2.4	26163	43605
2.5	23256	25840	2.6	17442	46512
2.7	14688	14688			

They are distinct, proving bare nonisomorphism. For a flag-transitive rank-two coset geometry, its two object counts and its number of incident pairs are $N/|P|$, $N/|Q|$, and $N/|P \cap Q|$. These are intrinsic even if an isomorphism does not preserve the specified action. We compare the two parabolic orders and their full intersection order for each of the three pairs in each of the twenty-one supplied rank-three families. Among all sixty-three comparisons, only active row 1, types $1, 3$, matches 2.6, and only active row 2, types $1, 3$, matches 2.3. All other candidate truncations are excluded by the intrinsic counts. The two matches are actual isomorphisms by the earlier literal coset maps, completing the assertion. □

The comparison row and its retained premises

The accepted named table of marks gives 137 conjugacy classes of subgroups and 71 564 248 subgroups of G. The supplied rank-three roster has nineteen bare incidence classes by the separate accepted flag-count and point-graph argument. Consequently the finite data give seven distinct rank-two geometries, two explicit truncations and nineteen distinct rank-three geometries. To promote these to the global nontruncation row

$$(\text{rank } 2, 3, 4, 5, 6) = (5, 19, 0, 0, 0),$$

we retain exactly the following inputs from L091 reference [9]: every eligible geometry appears in the supplied earlier classification, and there is no eligible geometry of rank at least four. Under these inputs, every possible rank-two truncation is among the sixty-three tested pairs, and no rank-three geometry can be a truncation of a higher rank one. The printed rank-three value seventeen is replaced by nineteen; the seventeen diagram labels of the supplement do not count its bare classes. The correction is explicit, while an independent global Janko census is not claimed.

The reference [9] original is outside the active 105 mathematical manuscripts, and its author supplement is an auxiliary source. No L097 recovery credit is awarded. This join does not import every metric, abstract subgroup name or automorphism assertion printed in that separate supplement. The new HS classification and the Hoffman–Singleton constructions of L091 have their own accepted proofs. Their integration, correction ledger and distinct whole-source review remain the final L091 admission steps. GAP and the complete table of marks retain their explicit software and classification premises. No new conjecture, Lean or compression result is claimed by this join.